

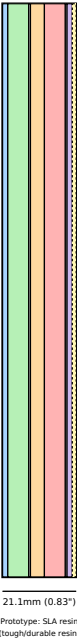
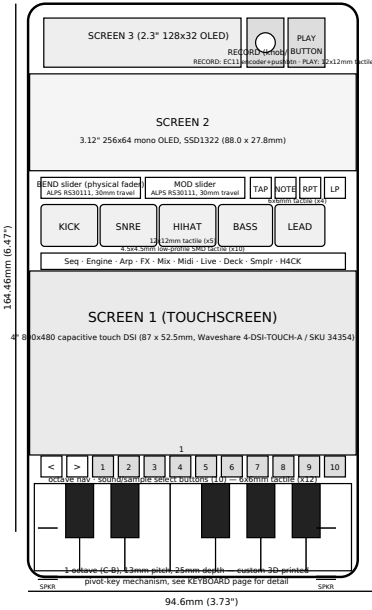
cy83rd3ck — TABLE OF CONTENTS

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Device: 94.6 x 164.46 x 21.1mm handheld music instrument. CM4 + Teensy 4.1, 3 screens, 1-octave printed pivot-key keyboard, 606090 LiPo. Status: CONCEPT — breadboard validation phase.

cy83rd3ck — MULTI-SCREEN ENCLOSURE CONCEPT (FROM SKETCH)



Layer legend (side-section, left=front / right=back):

- FRONT — 1.5mm front panel
- CTRL/DISPLAY — ~6mm (control PCB + display layer, incl. ~3mm for 16mm speakers)
- gap — 0.5mm assembly tolerance
- PCB — 4mm main board
- BATT — 6mm (606090 LiPo, ~4000mAh, ~2.1hr runtime)
- gap — 0.5mm: compressible foam pad (battery swelling allowance + anti-rattle)
- BACK — 1.5mm back panel
- TOL — 1.1mm assembly tolerance

Total: 21.1mm (0.83") — Prototype: SLA resin (tough/durable resin)

Layout summary (current concept):

- Screen 3 (top-left, 2.3" OLED): status/transport display, plus RECORD encoder (knob+pushbutton) and PLAY button.
- Screen 2 (3.12" OLED): waveform/secondary display.
- Row 1: BEND + MOD sliders (ALPS RS30111), plus TAP / NOTE / RPT / LP buttons.
- Row 2: 5 track-select buttons — KICK / SNRE / HIHAT / BASS / LEAD.
- Tab row: Seq/Engine/Arp/FX/Mix/Midi/Live/Deck/Smplr/H4CK — 10x low-profile tactile.
- Screen 1 (4" 800x480 DSI touch, 87x52.5mm): main UI — sequencer, sampler, deck, mixer, depending on active tab.
- Bottom: 1-octave custom 3D-printed pivot-key keyboard (7 white + 5 black, 13mm pitch, 12x FSR) — see KEYBOARD pages for mechanism detail.

cy83rd3ck	TITLE	Multi-Screen Enclosure Concept (3 displays + 2 OLEDs)	DWG NO.	CY83-ENC-010 (concept)
	BASIS	88.9 x 127.0mm (3.5 x 5.0in) front face, layout per hand sketch	SCALE	A5 NOTED
	STATUS	CONCEPT — part numbers / thicknesses / driver routing TBD	REV	A — June 2026

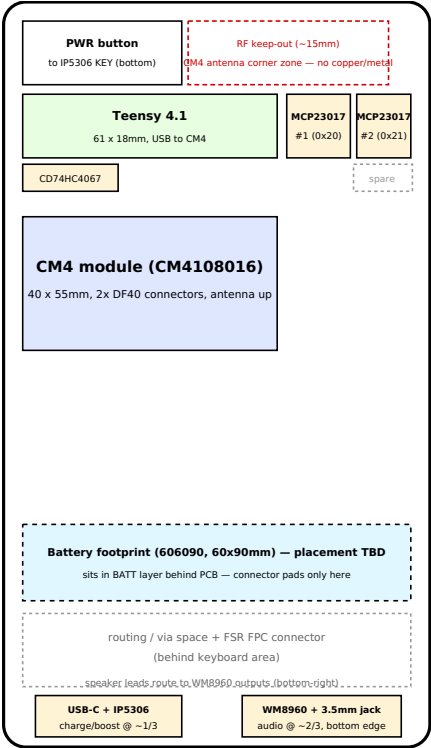
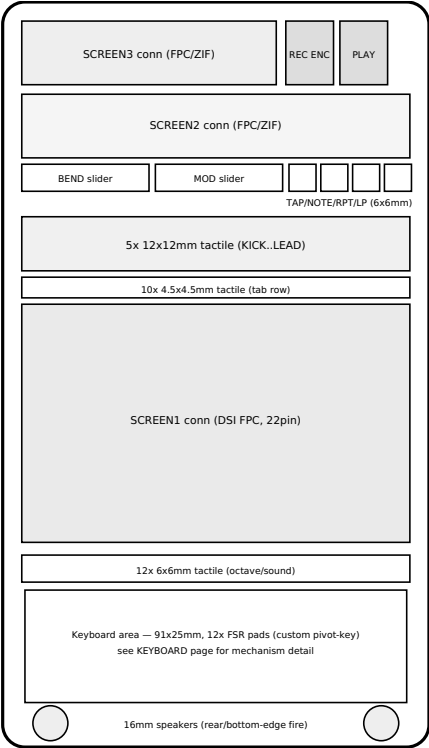
cy83rd3ck — BILL OF MATERIALS (PROTOTYPE)

Category	Component	Part / Spec	Notes
Compute	SBC	Raspberry Pi CM4108016 — 8GB RAM / 16GB eMMC / Wireless (\$252), 40x55mm module	Runs Chromium kiosk + cy83rd3ck app. Final device: bare module socketed onto main PCB; CM4IO Board (\$50.95) is breadboard/dev-only.
Compute	Input MCU	Teensy 4.1	USB MIDI controller for all physical inputs
Display	Screen 1	4" 800x480 capacitive touch DSI panel (87 x 52.5mm active area, Waveshare 4-DSI-TOUCH-A / SKU 34354, \$28.99)	Main touchscreen UI
Display	Screen 2	3.12" SSD1322 mono OLED, 256x64 (88.0 x 27.8mm)	Waveform display, SPI
Display	Screen 3	2.3" SSD1306 mono OLED, 128x32 (~56.6 x 14.1mm)	Status/transport, SPI/I2C
Controls	BEND / MOD sliders (x2)	ALPS RS30111, 10K linear, 30mm travel	Analog → Teensy ADC
Controls	TAP/NOTE/RPT/LP (x4)	6x6mm tactile switch	Direct GPIO
Controls	Track buttons (x5)	12x12mm tactile switch	Direct GPIO
Controls	Tab row (x10)	4.5x4.5mm low-profile SMD tactile	MCP23017 #1 (0x20)
Controls	Octave nav + sound select (x12)	6x6mm tactile switch	MCP23017 #2 (0x21)
Controls	RECORD encoder	EC11 rotary encoder w/ pushbutton	Direct GPIO (A/B/push)
Controls	PLAY button	12x12mm tactile switch	MCP23017 #1
Controls	Keyboard (1 oct, 7W+5B)	Custom 3D-printed pivot-key mechanism (SLA resin), 13mm white key pitch, 91x25mm keybed, 12x Interlink FSR — FSR 400 (7.62mm) preferred, pending nub-contact check at test print; FSR 402 (12.7mm) fallback needs 2-row stagger (risk #18). 1x 402 already in breadboard kit; 11 more to order. See KEYBOARD + FSR STRIP pages.	FSR matrix → CD74HC4067 mux → Teensy A0
Audio	Codec	WM8960 (I2S + I2C)	Stereo speaker amp + 3.5mm jack + mic in
Audio	Speakers (x2)	BeStar 16mm round, 0.7-1.5W, 4-8Ω (~2.5-3mm thick)	Rear/bottom-edge grilles
Power	Battery	606090 LiPo, ~4000mAh, 3.7V (6 x 60 x 90mm), ~\$8-12 — must include built-in protection circuit (PCM)	~2.1hr runtime @ ~1.2A avg draw; see CHARGE-WHILE-PLAYING page
Power	Power button	6x6mm tactile, top edge (sole top-edge item)	IP5306 KEY pin: short=on, double/long=off — see POWER BUTTON/GAUGE/SHUTDOWN page
Power	Fuel gauge	MAX17048 (~\$1), I2C addr 0x36	On CM4 I2C bus; battery % to app + Screen 3, drives safe-shutdown thresholds
Power	Charge/Boost	IP5306-based charge/boost, USB-C, 5V/2.1A	Powers CM4 + Teensy
Logic	GPIO expanders (x2)	MCP23017, addr 0x20/0x21, SSOP-28	I2C, shared bus w/ Teensy
Logic	Analog mux	CD74HC4067, SOIC-24	Keyboard FSR matrix
Connectors	USB-C	16-pin SMD mid-mount receptacle	PD charging
Connectors	3.5mm jack	PJ-320A-class slim SMD (~6mm depth)	WM8960 headphone out
Connectors	Screen 1 FPC	22-pin, 0.5mm pitch (CM4 DSI)	
Connectors	Screen 2/3 FPC + ZIF	0.5mm pitch FPC, matching ZIF sockets	
Connectors	Keyboard FSR link	14-pin 0.5mm FPC + ZIF socket (12 sig + 3.3V + GND)	
Enclosure	Case (front/back)	SLA resin, tough/durable resin, 1.5mm walls	21.1mm (0.83") total depth
Thermal	Spreader plate + pad	~24x18x1mm aluminum plate, flush in back panel window; 1-2mm silicone thermal pad to CM4 SoC	Passive cooling, see THERMAL MANAGEMENT page
Enclosure	Closure hardware	6x M2x6 screws + 6x M2x3 heat-set brass inserts (~\$2); insert iron tip ~\$10 one-time tool	Back-entry screws, lip-and-groove seam — see ENCLOSURE CLOSURE page
Enclosure	Foam sheet (thin, adhesive-backed)	~0.5mm soft PU/silicone foam sheet, ~\$3-5 — cut to size for: battery pad (60x90mm), Screen 1 gasket, FPC anti-rattle dots	Battery swelling allowance + anti-rattle, see risk log #22

Open items: final PCB layout integrating all of the above (see NET LIST page for full connectivity).

cy83rd3ck — PCB FLOORPLAN (CONCEPT, NOT ROUTED)

FRONT-FACING (switches, sliders, display connectors)BACK-SIDE (compute, logic ICs, power, connectors)



Notes: This is a single PCB with components on both sides — front side carries switch/slider footprints and display FPC/ZIF connectors aligned to the front-panel layout; back side carries the CM4 module socket, Teensy 4.1, audio codec, power management, GPIO expanders/mux, and the USB-C/3.5mm connectors (positioned at the bottom edge, centered between the speaker chambers, for enclosure cutout access). The CM4 module (40x55mm, ~4.7mm tall overall) mounts to the PCB back via 2x 100-pin Hirose DF40-series board-to-board connectors (the standard CM4 mating system — receptacles on our PCB, ~\$3 each; no standoffs needed at this stack height). It is placed behind the track-button row where there's a large open area; Teensy 4.1 (61x18mm) sits behind Screen 2. The battery sits in its own depth layer behind the PCB — only its connector pads are on the PCB, in the area behind Screen 1's lower portion. Keyboard FSR FPC connector sits at the board's bottom edge alongside USB-C/jack (see FSR STRIP WIRING and JACK PLACEMENT pages).

Next step for an actual routable board: import this floorplan into KiCad, place real footprints (CM4 castellated-edge socket, Teensy headers, SSOP/QFN/SOIC packages for the ICs, FPC/ZIF connectors, switch/slider THT or SMD footprints), then route power (5V/3.3V rails) and signal nets (SPI, I2C, GPIO matrix, analog) per the pin-wiring table.

cy83rd3ck — FRONT PANEL SHELL (OpenSCAD source)

First print target: a flat 1.5mm front panel (94.6 x 164.46mm) with all cutouts derived from the layout above. Keyboard area (91x25mm) houses a custom 3D-printed pivot-key mechanism — see [KEYBOARD](#) page. Open in OpenSCAD (free, opencad.org) to render and export STL for SLA printing. Use this to validate button/screen/keyboard alignment against sourced parts before modeling the full enclosure (walls, back, standoffs).

```
// =====
// cy83rd3ck - FRONT PANEL SHELL (concept, prototype)
// Overall: 94.6 x 164.46 x 1.5mm front panel, device depth 21.1mm (keyboard depth is PLACEHOLDER)
// Open this file in OpenSCAD (free, opencad.org) to view/export STL.
// All cutout positions derived from the front-panel diagram
// (cy83rd3ck_hardware_concept.pdf), 1 unit = 0.3534mm.
// =====

W = 94.6;
H = 164.46; // keyboard depth (25mm) per custom pivot-key mechanism (see KEYBOARD page)
T = 1.5;    // panel thickness
R = 7;      // corner radius
EXTRA = 4;  // extra height for cutout tools (through-cuts)

module rrect(w, h, r) {
  hull() {
    translate([r, r])      circle(r=r, $fn=48);
    translate([w-r, r])    circle(r=r, $fn=48);
    translate([r, h-r])    circle(r=r, $fn=48);
    translate([w-r, h-r])  circle(r=r, $fn=48);
  }
}

module panel() {
  linear_extrude(height=T) rrect(W, H, R);
}

module cut_rect(x, y, w, h) {
  translate([x, y, -EXTRA/2]) cube([w, h, T+EXTRA]);
}

module cut_circle(cx, cy, d) {
  translate([cx, cy, -EXTRA/2]) cylinder(d=d, h=T+EXTRA, $fn=48);
}

module front_panel() {
  difference() {
    panel();

    // ---- Display windows ----
    cut_rect(4.29, 79.61, 85, 50.5); // SCREEN1 cutout: active area (87x52.5) minus 1mm overlap/side, see SCREEN1 CUTOUT page
    cut_rect(8.91, 24.43, 76.778, 19.178); // SCREEN2 (3.12" SSD1322 OLED active area)
    cut_rect(4.51, 4.11, 56.6, 14.1); // SCREEN3 (2.3" 128x32 OLED)

    // ---- Record encoder + Play button (top row) ----
    cut_circle(68.11+3.5, 11.16+3.5, 7); // RECORD encoder shaft, d=7mm
    cut_rect(73.89, 5.16, 12, 12); // PLAY button

    // ---- TAP / NOTE / RPT / LP (6x6mm) ----
    cut_rect(63.80, 49.84, 6, 6);
    cut_rect(70.88, 49.84, 6, 6);
    cut_rect(77.97, 49.84, 6, 6);
    cut_rect(85.06, 49.84, 6, 6);

    // ---- Track buttons KICK/SNRE/HIHAT/BASS/LEAD (12x12mm) ----
    cut_rect(5.84, 57.63, 12, 12);
    cut_rect(22.80, 57.63, 12, 12);
    cut_rect(39.77, 57.63, 12, 12);
    cut_rect(56.38, 57.63, 12, 12);
    cut_rect(72.02, 57.63, 12, 12);

    // ---- Tab row x10 (4.5x4.5mm) ----
    for (i = [0:9]) {
      cut_rect(5.83 + i*8.726, 71.64, 4.5, 4.5);
    }

    // ---- Octave nav (<,>) + sound select x10 (6x6mm) ----
    oct_x = [3.71, 11.11, 18.50, 25.90, 33.30, 40.70, 48.10, 55.50, 62.89, 70.28, 77.69, 85.07];
    for (x = oct_x) {
      cut_rect(x, 129.77, 6, 6);
    }

    // ---- BEND / MOD slider slots (30mm travel x 4mm) ----
    cut_rect(2.98, 50.84, 30, 4);
    cut_rect(32.93, 50.84, 30, 4);

    // ---- Keyboard cutout (1 octave, FSR keybed pokes through) ----
    cut_rect(1.80, 137.69, 91.0, 25.0); // custom pivot-key Keybed, 12x FSR (see KEYBOARD page)

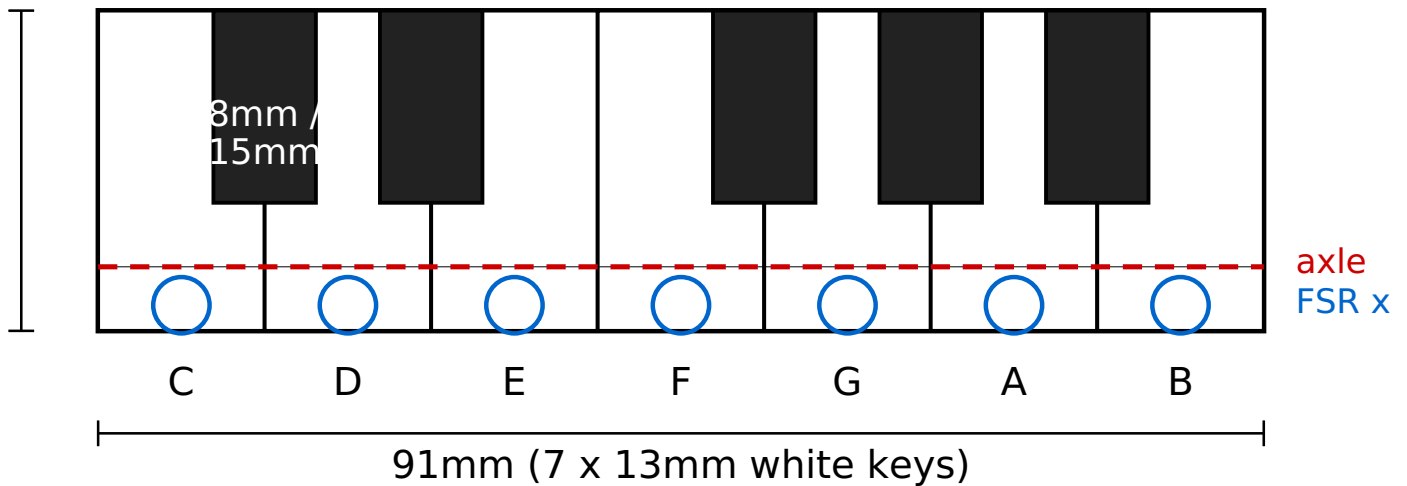
    // ---- Speaker grilles (bottom edge, simple slot vents) ----
    for (gx = [0:2]) {
      cut_rect(2.83 + gx*5, H-4, 3, 3); // left grille, 3 slots
      cut_rect(75.77 + gx*5, H-4, 3, 3); // right grille, 3 slots
    }
  }
}

front_panel();

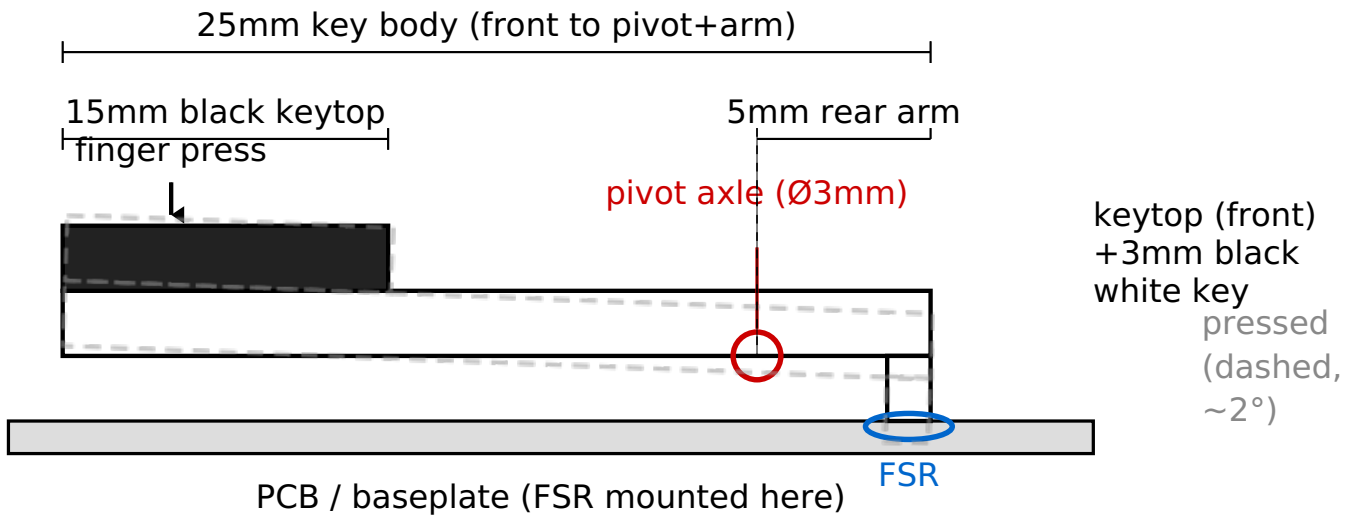
// =====
// NOTES:
// - This is the FRONT PANEL only (1.5mm) - a good first print to
//   validate button/screen/keyboard alignment against real parts.
// - Side walls + back panel + internal standoffs are NOT modeled
//   yet; this is a flat test plate.
// - BEND/MOD slots assume the slider lever protrudes through a
//   30mm slot; verify against the actual ALPS RS3011 lever travel.
// - Track button & octave/sound cutouts are sized to the SWITCH
//   (12x12 / 6x6mm), not a larger keycap - add keycap clearance
//   if using oversized caps.
// - Keyboard cutout is a single rectangle; the FSR keybed assembly
//   sits behind it. Dimensions per the custom pivot-key mechanism pages.
// =====
```

cy83rd3ck — KEYBOARD: PIVOT-KEY MECHANISM (CONCEPT)

TOP VIEW — 1 octave (7 white + 5 black), 13mm pitch, 91 x 25mm keybed



SIDE VIEW — single key, pivot/lever action (REST vs PRESSED)

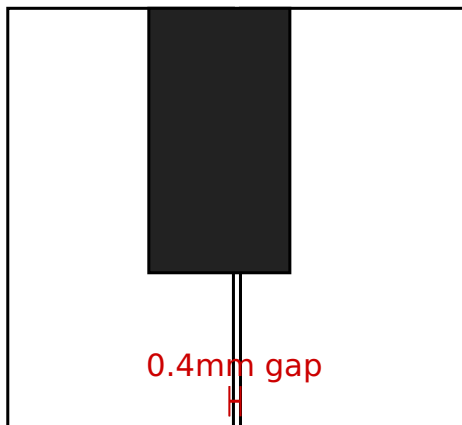


Mechanism notes:

- Each of the 12 keys (7 white @ 13mm pitch, 5 black @ 8mm wide x 15mm long) is a single printed lever pivoting on a shared Ø3mm axle running the width of the keybed.
- Lever ratio ~4:1 (20mm front arm : 5mm rear arm) — a ~1.5mm finger press at the key tip translates to ~0.4mm travel at the rear nub, enough to actuate an Interlink FSR per key (FSR 400, 7.62mm disc, preferred — see risk #18).
- White key body: 13 x 25 x 6mm. Black key: 8mm wide x 15mm long x +3mm raised on top of the white key's front portion.
- Axle sits ~5mm from the rear edge (20mm from the front tip). A small downward nub (3 x 6mm) on the rear arm contacts the FSR mounted on the PCB/baseplate below.
- Print orientation: SLA resin, keys printed individually or as a comb (all 12 in one piece with snap-in axle holes) — TBD based on first test print.
- 12x Interlink FSR required — FSR 400 preferred (risk #18); 1x 402 already in the breadboard kit, 11 more to order.
- This replaces the donor-keyboard approach; keybed footprint (91 x 25mm) and overall device height (164.46mm) from the placeholder resize remain valid for this mechanism.

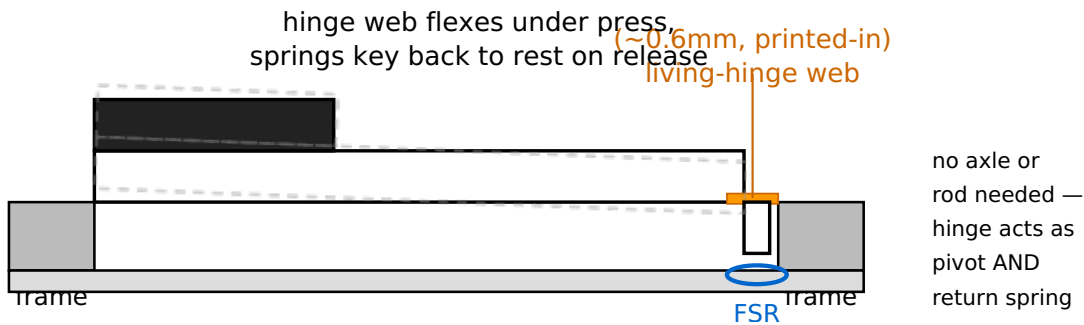
cy83rd3ck — KEYBOARD: GAPS, MOUNTING & RETURN MECHANISM (CONCEPT)

A — KEY-TO-KEY GAP (top view, zoomed on C/C# boundary)

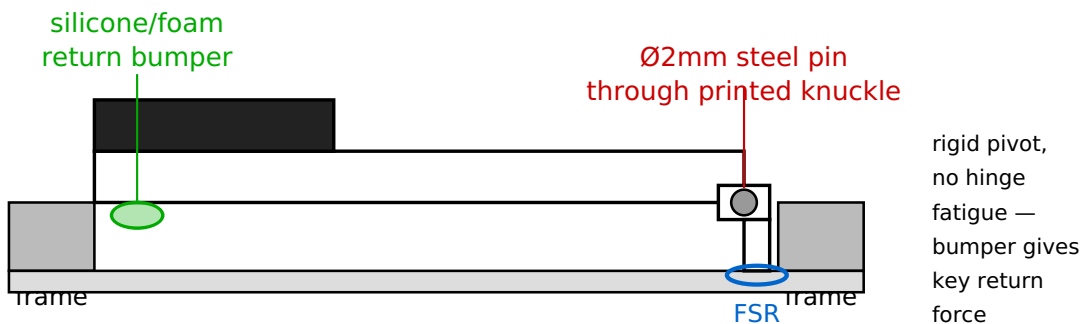


Uniform 0.4mm clearance on all key edges (white-white, white-black, black-black sides). Prevents binding under flex/rotation. Tune $\pm 0.1\text{mm}$ after first test print.

B — PRIMARY ASSEMBLY: LIVING-HINGE COMB (single SLA print)



C — FALLBACK: DISCRETE AXLE + RETURN BUMPER (if resin hinge proves too stiff/brittle)



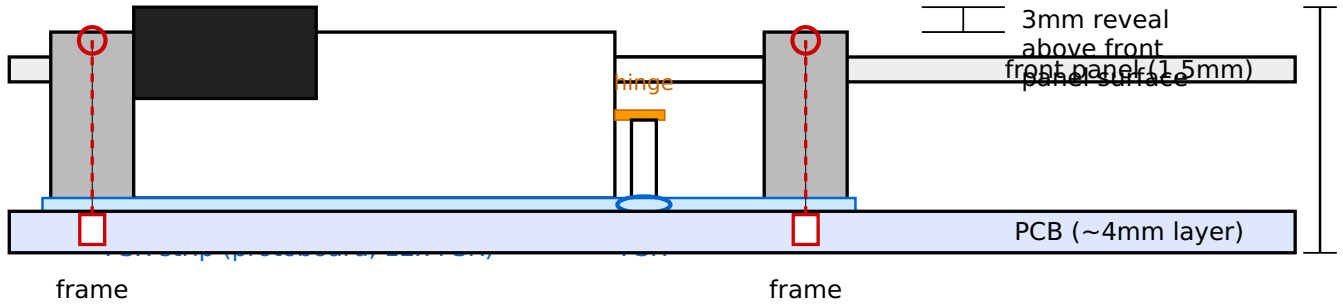
Assembly & mounting notes:

- **Gaps:** 0.4mm clearance on all key edges (white-white, white-black, black-black sides) — start here, tune $\pm 0.1\text{mm}$ after first test print based on resin shrinkage and fit.
- **Primary build (B):** all 12 keys + frame end-blocks printed as a single comb piece. A $\sim 0.6\text{mm}$ living-hinge web at the axle line (orange) acts as both the pivot and the return spring — press flexes the web, release springs the key back. Simplest option: one print, no fasteners, no separate axle hardware.
- **Fallback (C):** if the printed hinge proves too stiff (key won't move) or too brittle/fatigues over repeated presses, switch to a discrete $\text{Ø}2\text{mm}$ steel pin through printed knuckles for the pivot, plus a small silicone or foam bumper under the front tip of each key to provide return force. More parts, but more durable for heavy use.
- **Recommendation:** print one key (or a 2-3 key segment) of option B first. SLA resin living hinges can work but vary a lot by resin type/thickness — this is a fast, cheap test before committing to the full 12-key comb.
- **FSR mounting:** all 12 FSRs sit on a thin strip PCB or protoboard fixed to the baseplate, aligned under each key's rear nub — same FSR row as shown on the KEYBOARD mechanism page, wired through the CD74HC4067 mux to Teensy A0.

cy83rd3ck — KEYBOARD: MOUNTING / CROSS-SECTION (CONCEPT)

FULL STACK — front panel → key comb → FSR strip → PCB

black keytop (+3mm, proud of panel)



M2 screw,
frame end-block
→ PCB standoff
(both sides)

Mounting notes:

- The 12-key comb (keys + living-hinge + frame end-blocks, single SLA print) drops into the 91x25mm front-panel cutout from behind, with key tops passing through and sitting ~3mm proud of the front panel surface (matches the black keytop's raised height — white key tops sit flush with the panel face, black keys stand 3mm above).
- **Frame end-blocks** on each side of the comb extend down to the PCB layer and are secured with **M2 screws into printed/heat-set standoffs on the PCB** — two screws total (one per end-block) is the minimum; add a center support standoff if the comb flexes across its 91mm span.
- The **FSR strip** (thin protoboard with 12x Interlink FSRs — FSR 400 preferred, risk #18) is sandwiched between the rear nubs and the PCB, held in place by the same screw-down compression — no separate fasteners needed for the FSR strip itself, but alignment pins/slots may help keep it positioned during assembly.
- **0.4mm side gaps** between adjacent keys (from page 6) are maintained by the front-panel cutout edges constraining the comb's outer keys, and by the printed gaps within the comb itself for inner keys.
- **Stack budget — resolved:** the FSR strip is not a separate stacked layer — it's a thin (~0.4mm film, ~1mm thin protoboard/FPC) sensor pad mounted **directly on the main PCB**, within the PCB layer's existing 4mm component-height allotment (same as any other low-profile SMD part). With that reframing: key body (6mm) = CTRL/DISPLAY layer (6mm), PCB + FSR strip (~4mm) = PCB layer (4mm). Total = 10mm, matching the budget exactly, with the 1.1mm TOL margin elsewhere in the stack as normal slack. No enclosure depth change needed for this.
- **Wiring:** FSR strip leads route to the CD74HC4067 mux on the back side of the PCB (per the pin-wiring diagram), same as the BEND/MOD sliders.

cy83rd3ck — POWER BUDGET (ESTIMATE)

Component	Est. current @5V	Notes
CM4108016 (Chromium kiosk, WiFi on)	500-700 mA	Largest single load; varies with CPU/GPU activity
Screen 1 (4" DSI + backlight)	200-300 mA	Backlight dominates; dimmable
Teensy 4.1	~100 mA	Per PJRC spec at 600MHz
Screen 2 (SSD1322 OLED)	30-40 mA	OLED draw scales with lit pixels
Screen 3 (SSD1306 OLED)	15-20 mA	
WM8960 + 2x speakers (moderate volume)	100-200 mA	Peaks higher at max volume
Logic (2x MCP23017, mux, misc.)	10-20 mA	
TOTAL (typical)	~1.0-1.4 A	Peaks toward 1.6-1.8A possible

Battery runtime estimate:

- 606090 LiPo: 4000mAh @ 3.7V ≈ 14.8Wh.
- Through IP5306 boost to 5V at ~85% efficiency ≈ 12.6Wh ≈ ~2,520mAh @ 5V usable.
- At ~1.2A average draw: 2,520mAh / 1200mA ≈ ~2.1 hours of runtime.

This is a 4x improvement over the original 1000mAh LP523450 (~30min), at the cost of a small footprint/thickness increase (60x90 vs 34x50mm, 6mm vs 5.2mm — device depth 20.3mm → 21.1mm).

For longer runtime (3-4hr target), options remain:

- Larger custom-size pouch cell** (~80x130x6mm, ~7,000mAh by volume scaling) — fits the 94.6x164.46mm footprint with room to spare, similar thickness, but is a semi-custom part (longer lead time, confirm with supplier).
- Power management** — dim/sleep Screen 1's backlight when idle, throttle CM4 clock — could reduce average draw and extend the ~2.1hr figure.

This should be revisited once real current draw is measured on the breadboard — these are spec-sheet estimates, not measurements.

cy83rd3ck — PIN ASSIGNMENT TABLE (CONCEPT)

Raspberry Pi CM4 (via CM4IO Board 40-pin GPIO + DSI)

Interface	Pin(s)	Connects to
DSI	Dedicated DSI connector	Screen 1 (4" 800x480 touch panel)
I2C (audio bus)	GPIO2/3 (SDA/SCL)	WM8960 codec control
I2S	GPIO18/19/20/21 (PCM)	WM8960 audio data
USB	USB2.0 header	Teensy 4.1 (MIDI/serial)
microSD	—	Not used — CM4108016 has 16GB onboard eMMC
UART (optional)	GPIO14/15	Debug console

Teensy 4.1 (input controller)

Function	Pin(s)	Connects to
Keyboard FSR matrix (signal)	A0	CD74HC4067 mux SIG output
Mux select lines	4x digital (e.g. 2,3,4,5)	CD74HC4067 S0-S3
BEND slider	A1	ALPS RS30111 #1 wiper
MOD slider	A2	ALPS RS30111 #2 wiper
TAP / NOTE / RPT / LP	4x digital (e.g. 6-9)	6x6mm tactiles, direct GPIO
KICK / SNRE / HIHAT / BASS / LEAD	5x digital (e.g. 10-14)	12x12mm tactiles, direct GPIO
PLAY	1x digital (e.g. 15)	12x12mm tactile, direct GPIO
RECORD encoder A/B/push	3x digital (e.g. 16,17,22)	EC11 encoder
Tab row + octave/sound (22 buttons)	Wire2 (SDA2/SCL2, pins 25/24)	MCP23017 #1 (0x20) + #2 (0x21) — Teensy-owned bus, separate from CM4 I2C. (Wire2 not Wire1: pins 16/17 are reserved for analog A2/A3 — see NET LIST page.)
USB	USB device port	CM4 (MIDI/serial)

- Notes:**
- Pin numbers above are **placeholder assignments** — any free Teensy 4.1 digital/analog pin works for the direct-GPIO items; finalize once breadboard wiring begins.
 - I2C topology (resolved):** two independent I2C buses, no shared lines. **Teensy Wire2** (pins 24/25) talks to MCP23017 #1/#2 for input-side buttons. **CM4 GPIO2/3** talks to WM8960 only. Since these are separate controllers' buses entirely (not a shared bus with two masters), there's no conflict — just don't cross-wire Teensy's I2C pins to the CM4's I2C pins.
 - Screens 2 and 3 (SPI, mono OLEDs) — confirm whether driven by CM4 GPIO/SPI or by Teensy. Current circuit diagram shows them off the CM4 GPIO header; if so, CM4 also needs SPI0 (GPIO 19/21/23/24) wired to both displays with separate CS lines.

cy83rd3ck — BUILD SEQUENCE (BREADBOARD VALIDATION)

1. **Flash CM4108016** with Raspberry Pi OS, set up Chromium kiosk mode, install/load the cy83rd3ck web app. **eMMC wear protection (risk #23), part of this step:** disable swap (dphys-swapfile off), install log2ram (or mount /var/log as tmpfs), point Chromium's cache at tmpfs (--disk-cache-dir=/tmp/chromium-cache in the kiosk launch flags), and mount /tmp as tmpfs — the CM4's 16GB eMMC is non-replaceable, and Chromium's constant cache/log writes are the main wear source. Also enable the **overlay root filesystem** (raspi-config → Performance → Overlay FS) once the image is stable — makes hard power cuts harmless (risk #28); keep a small writable partition for app data.
2. **Mount CM4 on CM4IO Board**, connect Screen 1 via DSI. Verify display output, set rotation (480x800 panel → 800x480 landscape), confirm capacitive touch works.
3. **Breadboard Teensy 4.1:** wire BEND/MOD sliders to A1/A2. Verify analog readings via Arduino IDE serial monitor.
4. **Wire CD74HC4067 mux** + 1x Interlink FSR to Teensy A0. Verify FSR readings through the mux (single channel first).
5. **Wire 2x MCP23017** to Teensy's I2C bus. Breadboard the tab-row and octave/sound buttons; verify I2C reads.
6. **Wire direct-GPIO controls:** TAP/NOTE/RPT/LP, track buttons (KICK/SNRE/HIHAT/BASS/LEAD), PLAY, RECORD encoder.
7. **USB-connect Teensy ↔ CM4.** Verify MIDI messages from Teensy are received by the cy83rd3ck app.
8. **Wire WM8960 codec** to CM4 (I2S + I2C). Connect speakers + 3.5mm jack. Verify audio output and levels.
9. **Wire power management UX:** power tactile to IP5306 KEY pin (verify on/off behavior), MAX17048 fuel gauge to CM4 I2C (verify battery % readout via i2cget/Python). Hook gauge polling + low-battery thresholds into the Screen 2/3 sidecar service.
10. **Wire Screen 2 (SSD1322) and Screen 3 (SSD1306)** via SPI. Verify waveform and status displays render correctly.
11. **Print the key gap/hinge test coupon** (cy83rd3ck_key_gap_test.scad) — one white + one black finger on a living-hinge frame. Print at GAP=0.3/0.4/0.5mm (separate runs) and compare fit and hinge flex/durability under finger press.
12. **Iterate keyboard mechanism** based on test print — adjust hinge thickness, gaps, or fall back to discrete axle + bumper (option C) if needed.
13. **Print full 12-key comb**, wire the 12-FSR strip into the mux, verify full keyboard input.
14. **Full breadboard integration test** — all subsystems running together, end-to-end (keys → MIDI → app → audio out, displays updating).
15. **Measure real power draw** across the integrated breadboard; revisit POWER BUDGET page with actual numbers.
16. **Measure CM4 temps under load** (thermal camera or probe) on the breadboard; validate the THERMAL MANAGEMENT page assumptions (spreader plate + vent sizing).
17. **Design/print front panel + enclosure shell** around the validated breadboard layout.
18. **Enter the schematic in KiCad** from the NET LIST page (start the project from the official CM4IO reference design files), then begin placement/routing per the PCB LAYOUT PLAN page (4-layer, zones, corridors). Cross-check against breadboard learnings before layout.
19. **Transfer wiring to perfboard or PCB rev1**, using the PCB FLOORPLAN pages as a starting layout.
20. **Final assembly:** mount PCB, battery, speakers, keyboard comb, and the thermal spreader plate + pad (CM4 SoC → back panel window) into the enclosure.

cy83rd3ck — KNOWN RISKS & OPEN QUESTIONS

#	Item	Status / next step
1	Battery runtime — 4000mAh 606090 gives ~2.1hr at typical load (up from ~30min with 1000mAh)	See POWER BUDGET page. For 3-4hr, consider larger custom pouch cell (~80x130x6mm, ~7000mAh) or power management.
2	Living-hinge durability — untested SLA resin hinge for 12-key comb	Test-print 2-3 key segment first (build step 11). Fallback: discrete axle + bumper (KEYBOARD page 6, option C).
3	Keyboard mechanism stack height — resolved: FSR strip mounts directly on the main PCB (within its 4mm budget), not as a separate layer. Key body (6mm) + PCB w/ FSR (4mm) = 10mm, matches budget.	Resolved (concept) — see KEYBOARD MOUNTING/CROSS-SECTION page. Confirm with real FSR/protoboard thickness once parts are in hand.
4	I2C bus topology — Teensy reads MCP23017s, CM4 talks to WM8960	Resolved — two independent I2C buses (Teensy Wire2 for MCP23017s, CM4 GPIO2/3 for WM8960), no shared lines, no two-master conflict. See PIN ASSIGNMENT page.
5	FSR response for keys — FSRs are pressure sensors, not binary switches	Resolved (concept) — see FSR RESPONSE & VELOCITY MAPPING page: state machine (IDLE/ATTACK/HELD), slope-based velocity, per-key REST/MAX calibration, ≥500Hz scan target. Constants need tuning on real hardware.
6	Key-to-key gap (0.4mm) — may bind or rattle depending on resin shrinkage	Tooling ready — parametric test coupon (cy83rd3ck_key_gap_test.scad) prints one white + one black finger with adjustable GAP and HINGE.T. Print at 0.3/0.4/0.5mm and compare fit + hinge flex (build step 11).
7	Screen 1 cutout tolerance — 87x52.5mm active area behind front panel	Resolved (concept) — see SCREEN 1 CUTOUT & MOUNTING page: 85x50.5mm cutout (1mm overlap), 0.5x2mm step, M2 standoffs + foam gasket. Confirm standoff positions once panel is in hand.
8	Speaker acoustic chamber — 16mm speakers need a sealed/vented chamber	Resolved (concept) — see SPEAKER ACOUSTIC CHAMBER page: sealed ~Ø18x6mm chamber (~1.5cm³), 7x Ø2mm grille holes, rear-firing. Exact X/Y placement TBD pending battery/PCB layout.
9	Keyboard FSR strip wiring/routing — path from FSR strip to mux on PCB	Resolved (concept) — see FSR STRIP WIRING page: 14-pin FPC/ZIF, per-key voltage dividers, CD74HC4067 mux. Final trace path TBD at PCB rev1.
10	Microphone/SD card purchase — 64GB MakerDisk microSD ordered but CM4108016 has 16GB onboard eMMC	microSD likely unnecessary for final build; useful as a spare/dev tool — not a blocker.
11	Thermal — CM4 under sustained load in a sealed 21.1mm SLA resin enclosure, no vents/heatsink modeled	Resolved (concept) — see THERMAL MANAGEMENT page: flush aluminum spreader plate (thermal pad to SoC, primary path) + single mid vent row (upper row dropped by design — restore via top-edge slots if temps run hot). Validate with real temps once breadboard is under load.
12	CM4/Teensy board mounting — no defined standoff/mounting layout for the CM4+CM4IO or Teensy 4.1 in the enclosure	Resolved (concept) — both are PCB-mounted (CM4 via 2x 100-pin Hirose DF40 board-to-board receptacles — the standard CM4 mating connectors; Teensy via headers/solder), no standoffs needed. Positions on PCB FLOORPLAN back-side page; CM4 module height (~4.7mm) fits PCB layer + tolerance gap. CM4IO Board clarified as breadboard-only.
13	USB-C / 3.5mm jack placement — connector types specified in BOM, but physical position/edge on the enclosure not shown on any diagram	Resolved (concept) — see USB-C & 3.5MM JACK PLACEMENT page: bottom edge (centered between speaker chambers), USB-C mid-mount in PCB notch (9.5x3.5mm cutout), 3.5mm jack on PCB surface (Ø6.5mm cutout). Fine X/Z coords at PCB rev1.
14	Charge-while-playing thermal/safety — CM4 load + IP5306 charging current simultaneously in an enclosed case	Resolved (concept) — see CHARGE-WHILE-PLAYING page: IP5306 pass-through is designed-for behavior (~1-1.5W worst-case heat); mitigations = top-edge placement near vents, copper pour, PCM-protected cell, battery in separate layer. Validate temps while charging at build steps 15-16.
15	DSI cable routing — Screen1's DSI FPC must route from panel to CM4 within the 21.1mm stack without being pinched by the keyboard mechanism or battery	Resolved (concept) — see DSI CABLE ROUTING page: ZIF placed adjacent to panel's FPC exit, single gentle fold (~15-25mm run), never crosses keyboard/battery. ZIF coordinate locks at PCB rev1; confirm FPC exit orientation when panel arrives.
16	WiFi antenna placement — CM4 wireless module needs clearance; sealed resin case with nearby battery/speakers/FSRs could affect signal	Resolved (concept) — see WIFI/BT ANTENNA page: resin is RF-transparent; orient CM4 antenna corner toward top edge, ~15mm RF keep-out (no copper/metal/spreader), U.FL flex antenna as fallback. RSSI sanity check at build step 14.
17	Firmware/software integration — Teensy USB MIDI firmware ↔ cy83rd3ck web app integration layer not yet documented	Resolved (concept) — see FIRMWARE/SOFTWARE INTEGRATION page: class-compliant Teensy USB MIDI → Chromium Web MIDI API → existing app engine; full MIDI message map drafted. Largest unknown = Screen 2/3 Python sidecar; prototype early (needs only a Pi + one OLED).
18	11x FSR sourcing/lead time — significant cost driver (~\$77, COST SUMMARY page); could be a schedule bottleneck for build steps 11-13 if special-order	Resolved + new finding — Interlink FSR 400-series is standard stock at DigiKey/Mouser (no lead-time risk: 402 ~\$7-8, 400 ~\$4-5). BUT: FSR 402 disc is 18.3mm — too wide for single-row 6.5mm spacing. Preferred fix: switch to FSR 400 (7.62mm disc, fits, cheaper — drops FSR line item ~\$77→~\$50). See FSR STRIP WIRING page note. Confirm nub contact on FSR 400 active area at test print.
19	Enclosure closure method — how front/back shells join (screws into bosses, snap-fit, or adhesive) is undefined; affects serviceability, tolerance stack, and internal space for bosses	Resolved (concept) — see ENCLOSURE CLOSURE page: 6x M2 screws from the back into heat-set brass inserts, boss columns in the perimeter wall, lip-and-groove seam. No footprint/depth change. Boss coordinates lock at build step 17 CAD work.
20	Audio noise from boost converter — IP5306 switching noise shares power rails with the WM8960 codec; whine/hiss bleeding into the audio path is a classic failure in this device class	Resolved (concept) — see AUDIO POWER NOISE ISOLATION page: rail split + ferrite/bulk caps + dedicated LDO for WM8960 analog supply, plus PCB layout rules (separation, ground returns). Listen test at build step 8.
21	OLED burn-in — Screens 2/3 show mostly static status content; mono OLEDs burn in within months of always-on static pixels	Resolved (concept) — four software mitigations specified in the Screen 2/3 sidecar notes (FIRMWARE/SOFTWARE INTEGRATION page): pixel shifting, ~50-60% default contrast, idle screen-off, no static 100%-on regions. Build into the sidecar from day one.
22	Battery swelling clearance — LiPo pouches swell with age; a 606090 sandwiched tight between PCB and back panel can bulge the case or stress the cell	Resolved (concept) — the existing 0.5mm battery-to-back-panel gap is now specified as a compressible foam pad (~0.5mm soft PU/silicone foam, 60x90mm): holds the cell snug normally, compresses if it swells, with the 1.1mm TOL layer as further headroom. Battery fixed to PCB side with double-sided tape (not to back panel, so the back shell removes freely). Inspect cell for swelling whenever the case is open; PCM-protected cell already required (risk #14). No depth change.
23	eMMC wear — Chromium writes cache/logs constantly; CM4 eMMC is non-replaceable	Resolved (concept) — concrete config folded into build step 1: swap off, log2ram/tmpfs logs, Chromium cache on tmpfs (- - disk-cache -d1r) /tmp on tmpfs. App state (IndexedDB/localStorage) still persists to eMMC — that is fine, it is low-volume; only high-churn cache/log writes are redirected.
24	ESD on exposed inputs — fingers on FSR keys and touch panel are ESD paths into Teensy/CM4 ADC lines	Resolved (concept) — TVS diode arrays (e.g. 3x TPD4E05, 4ch each = 12 FSR lines, ~\$0.50 total) placed at the FSR FPC connector on the main PCB, clamping spikes at the entry point before they reach the Teensy ADC. Touch panel has its own controller protection; sliders/buttons behind printed caps are lower risk. PCB rev1 schematic item — no breadboard action needed.
25	Resin aging/durability — standard SLA resin yellows in UV and embrittles over months	Partially addressed (BOM already specifies tough/durable resin). Add a drop test before any unit leaves the bench; consider UV-resistant resin or paint for longer-lived units.
26	Key lateral wobble — living hinge constrains pitch rotation but not side-to-side twist on a 25mm lever; keys may feel sloppy	Test coupon will reveal severity (build step 11). If present: add a thin guide rib under each key slot in the front-panel cutout.
27	Power on/off button — no power switch existed in the design	Resolved (concept) — see POWER BUTTON/GAUGE/SHUTDOWN page: 6x6mm tactile on top edge (sole top-edge item), wired to IP5306 KEY pin (short=on, double/long=off). Cutout added at build step 17 CAD work.
28	Safe shutdown — hard power cuts risk corrupting the CM4 eMMC filesystem	Resolved (concept) — two layers (see POWER page): overlay root FS makes any hard cut harmless; gauge-driven graceful shutdown (~8% warn, ~5% clean shutdown) handles battery-flat. Button power-off = hard cut, safe via layer 1.
29	Battery gauge — basic IP5306 has no I2C telemetry	Resolved (concept) — MAX17048 fuel gauge (~\$1) on the CM4 I2C bus (addr 0x36, coexists with WM8960); sidecar service polls it and feeds the app battery icon + Screen 3 + the shutdown thresholds. See POWER page.
30	No RTC — CM4 lacks a battery-backed clock; system time is wrong until WiFi/NTP connects	Noted, low priority. Mostly cosmetic for a music device (file timestamps). Fix if it matters: RTC module (DS3231, ~\$2) on the CM4 I2C bus + coin cell.
31	Regulatory compliance (FCC/CE) — irrelevant for the personal prototype, but production units (Crowd Supply) containing an intentional radiator (WiFi/BT) require certification	Noted for later. CM4's wireless is pre-certified as a module, which substantially reduces (but does not eliminate) the testing burden. Budget/schedule item for any production decision — not a prototype concern.
32	Overlay-FS update workflow — with root read-only, updating the kiosk image or web app requires disabling the overlay, updating, re-enabling	Noted (workflow, not hardware). Document the 3-step update procedure in the repo README when overlay FS is enabled at build step 1, so future updates aren't confusing.
33	Top-edge real estate — USB-C, 3.5mm jack, power button, and vent slots originally crowded the top edge	Resolved by relocation — USB-C + 3.5mm moved to the bottom edge (centered between speaker chambers), power button is the sole top-edge item, vents are on the back face (THERMAL page). Bottom edge check: speakers ~25mm each corner + both connectors in the central ~45mm — fits. Confirm in the build-step-17 edge elevations.

Risk log status: complete for the concept phase (33 items). Remaining unknowns are by design discoverable only through physical testing — hinge/wobble (test coupon), FSR feel (steps 11-13), power/temps (steps 15-16), audio noise (step 8), WiFi (step 14). New items should be added as the build surfaces them.

cy83rd3ck — COST SUMMARY (PROTOTYPE BOM)

CONFIRMED (already purchased for breadboard kit)

Item	Cost
Raspberry Pi CM4108016 (8GB/16GB/Wireless)	\$252.00
Raspberry Pi CM4IO Board	\$50.95
Teensy 4.1 (pins soldered)	\$48.00
Screen 1 — Waveshare 4-DSI-TOUCH-A (SKU 34354)	\$28.99
Interlink 0.5" circular FSR (x1, from breadboard kit)	~\$13.21*
Subtotal (per-unit BOM, confirmed)	~\$393.15

*Exact FSR unit price not separately visible in cart screenshot — using the "Alligator Clip with Pigtail" line price as a placeholder reference; confirm against actual receipt.

ESTIMATED (not yet purchased — rough market prices, confirm before ordering)

Item	Est. cost
Screen 2 — 3.12" SSD1322 OLED	~\$15
Screen 3 — 2.3" SSD1306 OLED	~\$8
ALPS RS30111 sliders (x2)	~\$8
WM8960 codec breakout	~\$10
BeStar 16mm speakers (x2)	~\$4
606090 LiPo battery (~4000mAh)	~\$8-12
IP5306 charge/boost breakout	~\$4
MCP23017 (x2)	~\$3
CD74HC4067 mux	~\$2.50
Interlink FSR x11 more needed (FSR 400 preferred @ ~\$4-5; FSR 402 @ ~\$7-8 needs 2-row stagger)	~\$50-77
Misc. tactiles, encoder, connectors, FPCs	~\$15
SLA resin (enclosure + keyboard comb prints)	~\$15
Aluminum spreader plate + thermal pad	~\$3
M2 screws + heat-set inserts (closure)	~\$2
Subtotal (estimated)	~\$174.50

ONE-TIME TOOLS (not part of per-unit BOM)

Item	Cost
Breadboard, jumper wires, alligator clips, resistors	~\$32.31
Digital multimeter kit	\$24.27
Soldering kit	~price TBD (cart cut off)
Heat-set insert tip for soldering iron	~\$10
Subtotal (tools)	~\$56.58+

Estimated grand total (per-unit BOM): ~\$393 (confirmed) + ~\$174.50 (estimated) ≈ **~\$565-570**
Plus ~\$57+ in one-time tools (not repeated for future units).

Notes:

- These numbers are rough — replace estimates with real prices as parts are sourced.
- The 11x additional FSRs (~\$77) are a notable cost driver from the keyboard mechanism decision — worth keeping an eye on, or sourcing in bulk if a price break is available.
- This is prototype/single-unit cost — production cost at volume (for Crowd Supply) would differ significantly (lower per-unit part costs at scale, but add PCB fab/assembly, enclosure tooling/printing, packaging).

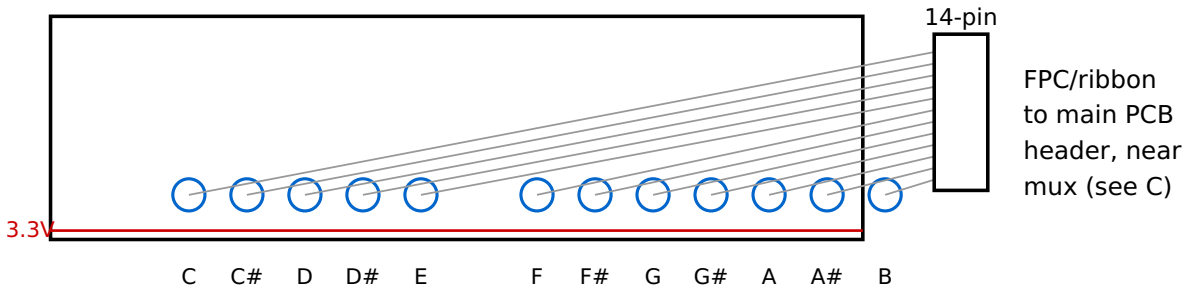
cy83rd3ck — REVISION HISTORY

Date	Change
Jun 2026	Initial enclosure concept: 88.9 x 127mm, 14-white/10-black key layout, 3-screen design.
Jun 2026	Switches/sliders sized to real parts; device grew to 94.6 x 133.24mm to fit 6x6/12x12mm tactiles and 4.5x4.5mm tab-row buttons.
Jun 2026	Side-section depth finalized at 20.3mm (0.8") with LP523450 battery + 16mm speakers.
Jun 2026	Battery upgraded: LP523450 (1000mAh, ~30min runtime) → 606090 (4000mAh, ~2.1hr runtime). Device depth 20.3mm → 21.1mm.
Jun 2026	CM4 variant changed: CM4104032 → CM4108016 (8GB/16GB/Wireless) due to stock availability.
Jun 2026	CM4 IO board selected: official Raspberry Pi CM4IO Board (\$50.95) over Waveshare alternative — cheaper, KiCad reference files available.
Jun 2026	Screen 1 resized: corrected to true 800x480 aspect ratio (88 x 52.8mm), device height grew 133.24mm → 151.51mm.
Jun 2026	Screen 1 part finalized: Waveshare 4-DSI-TOUCH-A (SKU 34354, \$28.99) — same active area, better bezel, cheaper than original 21687.
Jun 2026	Keyboard donor search (Raimy 37-key, EOOLEOW, Kikkerland keychain) — none matched required pitch + raised black keys.
Jun 2026	Decision: custom 3D-printed pivot-key keyboard mechanism (13mm pitch, 25mm depth). Device height grew 151.51mm → 164.46mm.
Jun 2026	Keyboard mechanism detailed: pivot/lever action, living-hinge comb (primary) + discrete axle/bumper (fallback), gap tolerances, mounting cross-section into device stack.
Jun 2026	Added power budget, pin assignment table, build sequence, risk log, and cost summary documentation.

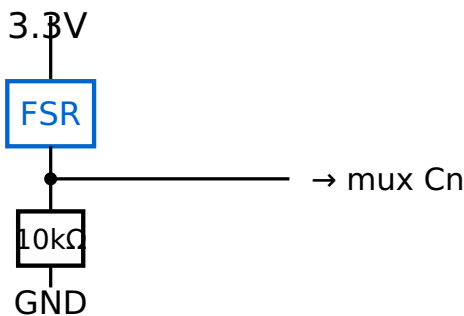
Status remains **CONCEPT** — breadboard validation in progress. This document will continue to evolve as parts are sourced, tested, and the design moves toward a routable PCB.

cy83rd3ck — KEYBOARD: FSR STRIP WIRING & ROUTING (CONCEPT)

A — FSR STRIP LAYOUT (top view, 91 x 25mm keybed)



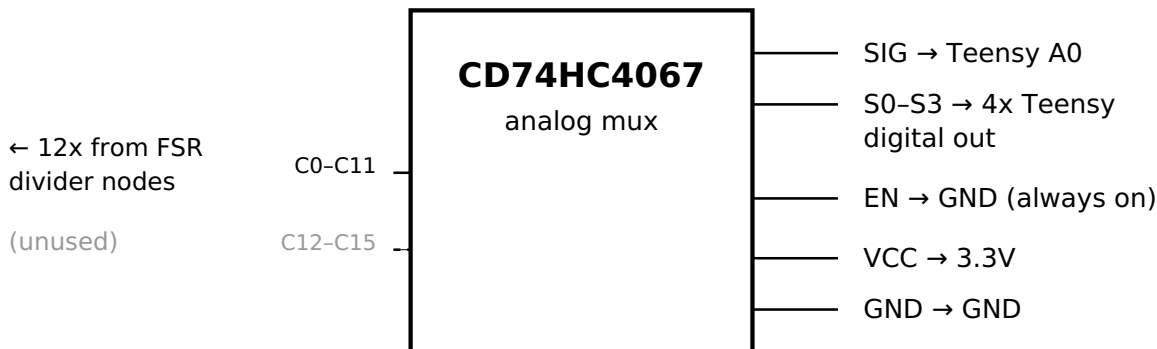
B — PER-KEY VOLTAGE DIVIDER (x12, identical)



x12 — one per key, all sharing the 3.3V and GND rails on the FSR strip protoboard.

Each divider node feeds one CD74HC4067 channel (C0-C11).

C — CD74HC4067 MUX → TEENSY 4.1

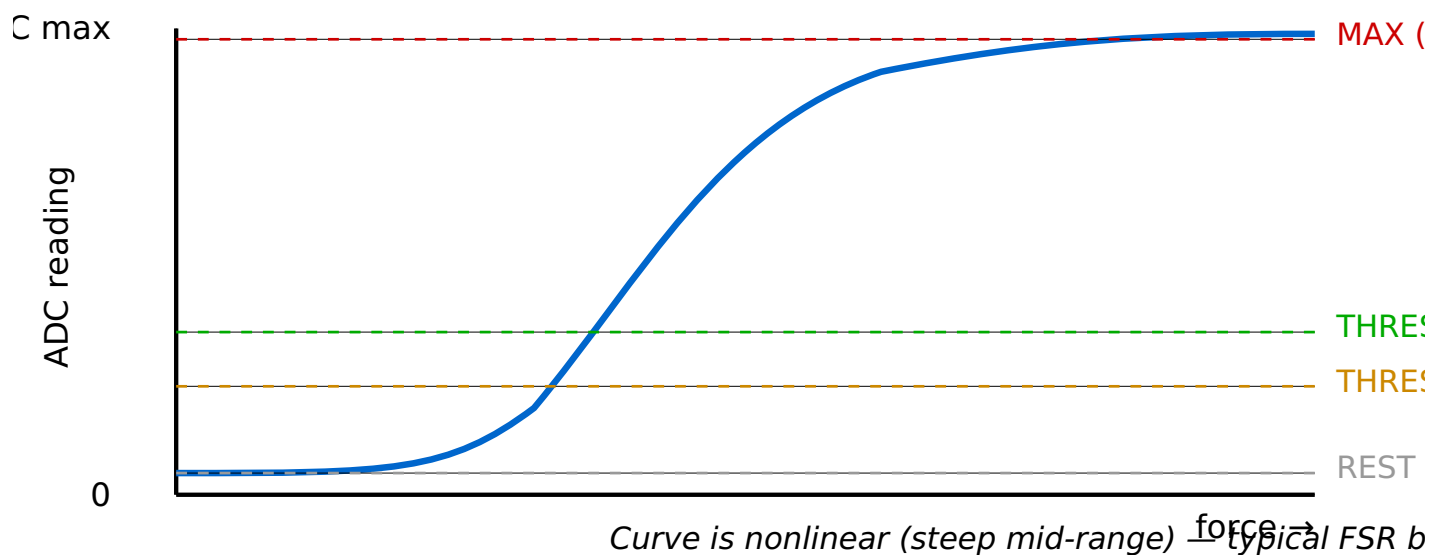


Routing notes:

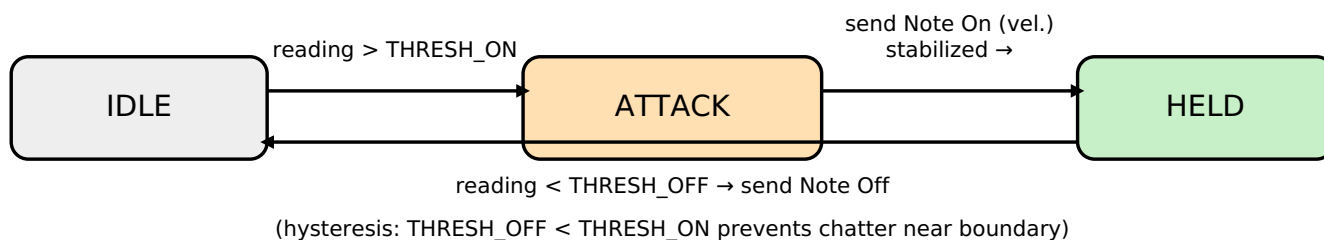
- **FSR strip:** a thin protoboard runs the 91mm width of the keybed, holding all 12 FSRs at the rear-nub positions shown in (A) — 7 under white keys, 5 under black keys, matching the KEYBOARD mechanism page layout. A shared 3.3V rail and GND rail run along the strip.
- **▲ Sensor size constraint (added after part lookup):** the Interlink FSR 402 (12.7mm active area) has an **18.3mm overall disc diameter** — too wide for the ~6.5mm single-row spacing drawn in (A). Two solutions: **(preferred) switch to the smaller Interlink FSR 400** — 7.62mm disc, fits a single row at this pitch, lower cost (~\$4-5 each), same electrical behavior — and size the rear nubs (~3x6mm) to its smaller active area; or keep the 402 and **stagger the FSRs in two rows** (white-key row rear, black-key row forward, with black keys' rear nubs at a different Y) — more mechanical complexity. Update diagram (A) and the BOM once the choice is confirmed against nub-contact reliability in the test print.
- **Per-key divider (B):** each FSR forms a voltage divider with a 10kΩ pulldown to GND. The divider node (FSR/resistor junction) is the analog signal for that key, routed to one CD74HC4067 channel (C0-C11).
- **Connector:** a 14-conductor connection (12 signal + 3.3V + GND) links the FSR strip to the main PCB. Given the tight 21.1mm stack, a **14-pin 0.5mm FPC + ZIF socket** (same style as the Screen2/3 connectors) is preferred over a discrete pin header/ribbon for lower profile. **ESD protection (risk #24):** place TVS diode arrays (3x 4-channel, e.g. TPD4E05) directly at this connector's landing on the main PCB — clamps finger-borne static at the entry point before it reaches the mux/Teensy ADC.
- **Mux placement (C):** the CD74HC4067 lives on the main PCB's back side, in the Logic section near the MCP23017s (per PCB FLOORPLAN page). The FSR strip's FPC routes from the keybed area, under/around the battery, to this header — exact trace path TBD once PCB layout begins, but the connector should be placed on the PCB edge closest to the keybed to minimize routing distance.
- **Channels C12-C15** are unused (12 of 16 channels populated) — could be reserved for future expansion (e.g. aftertouch, additional sensors) without hardware changes.
- **Mux control:** SIG → Teensy A0 (single ADC pin reads all 12 keys via mux scanning), S0-S3 → 4 Teensy digital outputs select the active channel, EN tied to GND (mux always enabled), VCC/GND shared with Teensy's 3.3V/GND.

cy83rd3ck — KEYBOARD: FSR RESPONSE & NOTE-ON/VELOCITY MAPPING (CONCEPT)

A — DIVIDER RESPONSE CURVE (ADC reading vs. key press force)



B — PER-KEY STATE MACHINE



Velocity mapping (in ATTACK state):

- While in ATTACK, sample the reading at the per-key scan rate and track the **slope** ($\Delta \text{reading} / \Delta \text{time}$) from THRESH_ON until the reading stabilizes (e.g. slope drops below a small epsilon, or a fixed 5-10ms attack window elapses).
- Map the measured slope to MIDI velocity: $\text{velocity} = \text{constrain}(\text{map}(\text{slope}, \text{SLOPE_MIN}, \text{SLOPE_MAX}, 1, 127), 1, 127)$ — a faster rise (hard press) → higher velocity; a slow rise (gentle press) → lower velocity.
- SLOPE_MIN/SLOPE_MAX are tuned empirically once real FSRs are on the bench — expect to need per-key or global calibration.

Per-key calibration:

- At startup (or a dedicated calibration mode), record each key's **REST** baseline (no press) — accounts for any mechanical preload from the pivot/nub resting on the FSR.
- Optionally guide the user through pressing each key fully to record **MAX**.
- Normalize: $\text{norm} = \text{constrain}((\text{reading} - \text{REST}) / (\text{MAX} - \text{REST}), 0, 1)$ — define THRESH_ON/OFF in normalized terms (e.g. 0.10 / 0.05) so thresholds are consistent across keys despite FSR-to-FSR and mechanical tolerance.

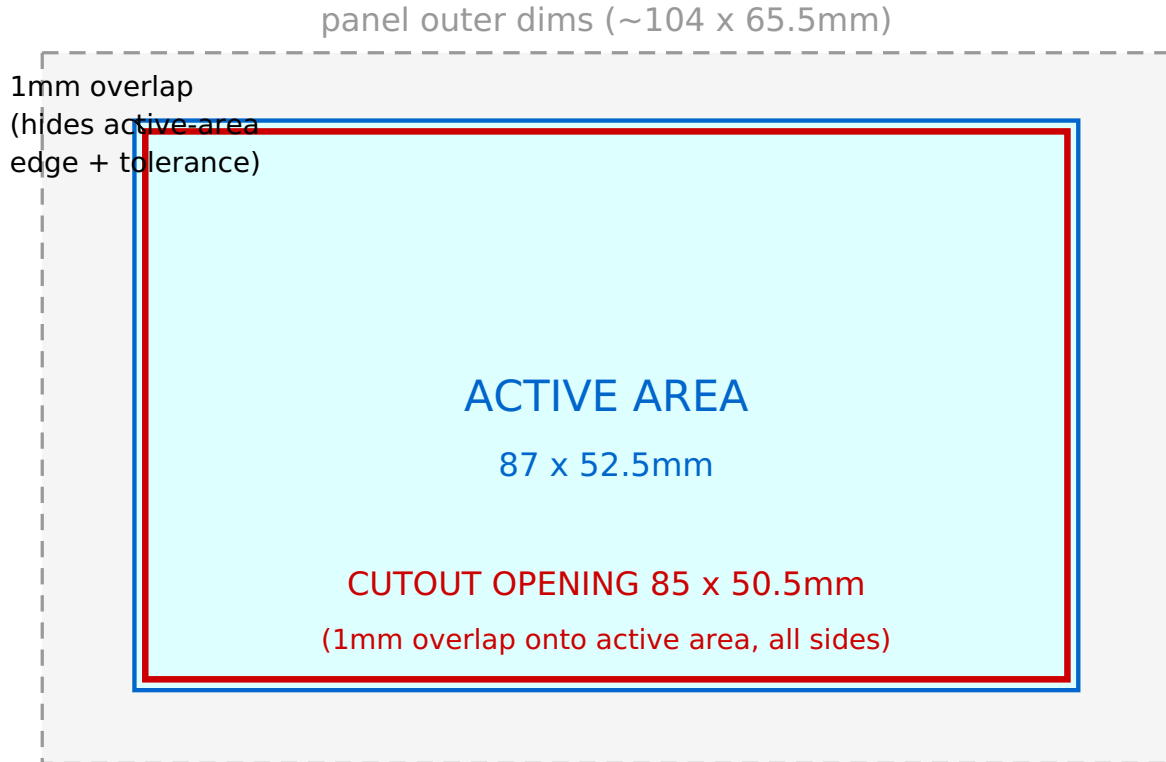
Scan rate / timing:

- 12 keys via the CD74HC4067 mux on Teensy A0. Mux channel-switch settling time is ~5-10µs; Teensy 4.1's ADC conversion is low-µs — a full 12-channel scan cycle in well under 1ms is easily achievable.
- Target: **≥500Hz effective scan rate per key** (full cycle ≤2ms) for responsive note-on detection and meaningful velocity slope measurement. Teensy 4.1 has ample headroom for this alongside its other duties (sliders, buttons, USB MIDI).

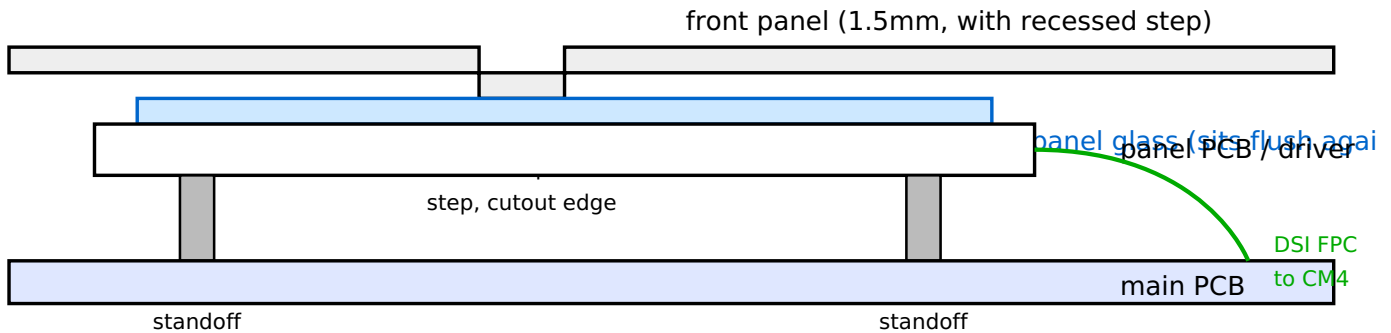
Open follow-up: all threshold/slope constants above are placeholders — finalize once real FSRs + the printed pivot-key mechanism are on the bench and actual ADC response curves can be measured (build steps 11-13 on the BUILD SEQUENCE page).

cy83rd3ck — SCREEN 1: CUTOUT TOLERANCE & MOUNTING (CONCEPT)

A — FRONT VIEW: cutout vs. active area vs. panel outer dims



B — CROSS-SECTION: stepped recess & mounting



Cutout sizing:

- Active area: **87 x 52.5mm** (Waveshare 4-DSI-TOUCH-A / SKU 34354). Panel outer dims: ~**104 x 65.5mm** (full PCB/driver board footprint).
- Front panel **cutout opening: 85 x 50.5mm** — **1mm smaller than the active area on all sides**. This means the front panel slightly overlaps the outer edge of the active area, which:
 - hides any inactive border pixels or slight active-area-to-glass-edge variance,
 - absorbs ± 0.3 - 0.5 mm SLA print tolerance without ever revealing the panel's bezel or mounting holes,
 - gives a clean, even-looking window regardless of the panel's own (uneven) bezel — consistent with the "hide the bezel behind the panel" decision made earlier.

Stepped recess (B): a **0.5mm deep x 2mm wide step** runs around the inside edge of the cutout. The touch panel's front glass sits flush against this step, which:

- sets the panel's z-position precisely (glass surface ends up ~0.5mm below the front panel's outer face — slight recess protects the glass edge),
- provides a registration surface so the panel can't shift during assembly.

Mounting: the panel's PCB/driver board is secured to the main PCB via **2x M2 standoffs** (one per side, more if the panel's mounting holes allow) — same approach as the keyboard comb and CM4IO board. The standoff height is set so the glass presses gently against the front-panel step (slight spring/compression from a thin foam gasket between glass and step is recommended to absorb tolerance stack-up).

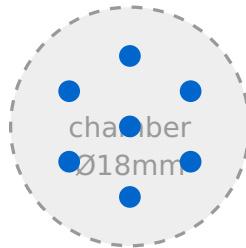
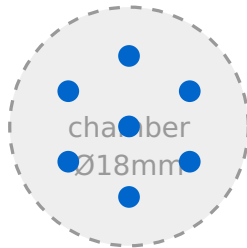
Open follow-up: exact panel PCB mounting hole positions aren't in the datasheet yet — confirm standoff positions once the panel is physically in hand (it's on order). The 85x50.5mm cutout and 1mm overlap should not need to change regardless of mounting hole locations, since those are independent of the active-area/cutout relationship.

cy83rd3ck — SPEAKER ACOUSTIC CHAMBER (CONCEPT)

A — BACK PANEL VIEW: chamber + grille placement (bottom corners)

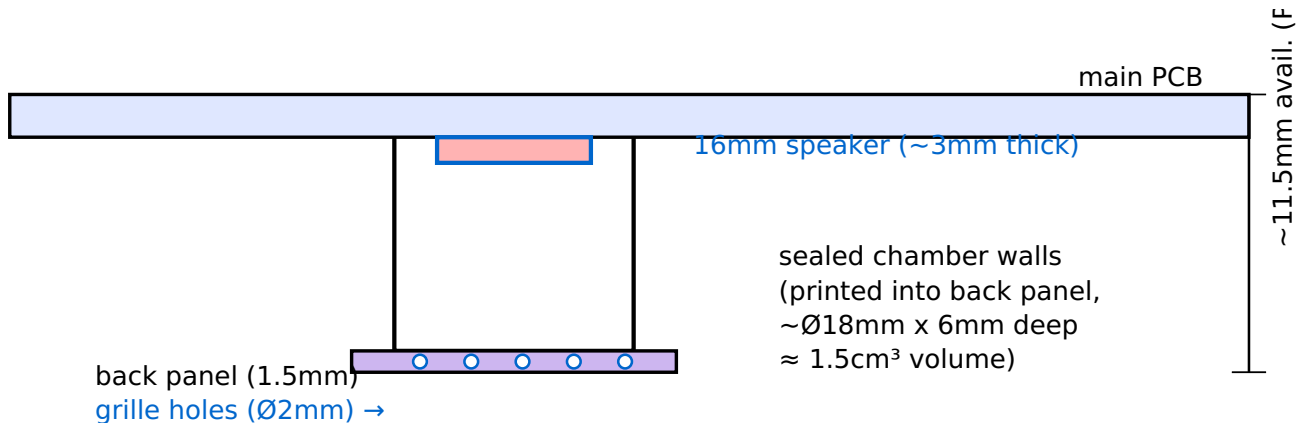
back panel (bottom portion shown)

keyboard mechanism area (front layer, above)



7x Ø2mm grille
holes per speaker
(~22mm² open
area each)

B — CROSS-SECTION: sealed chamber behind one speaker

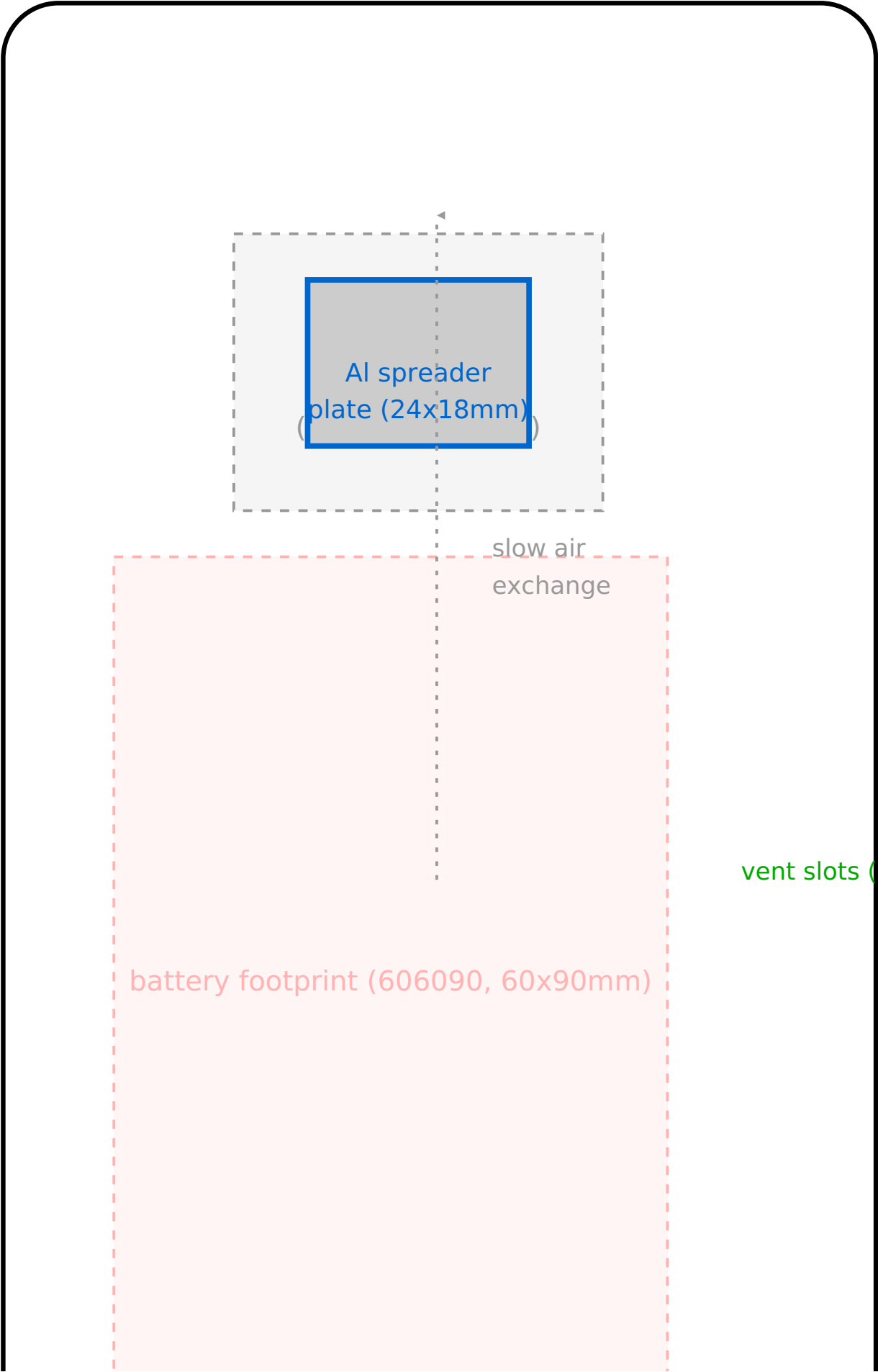


Chamber design:

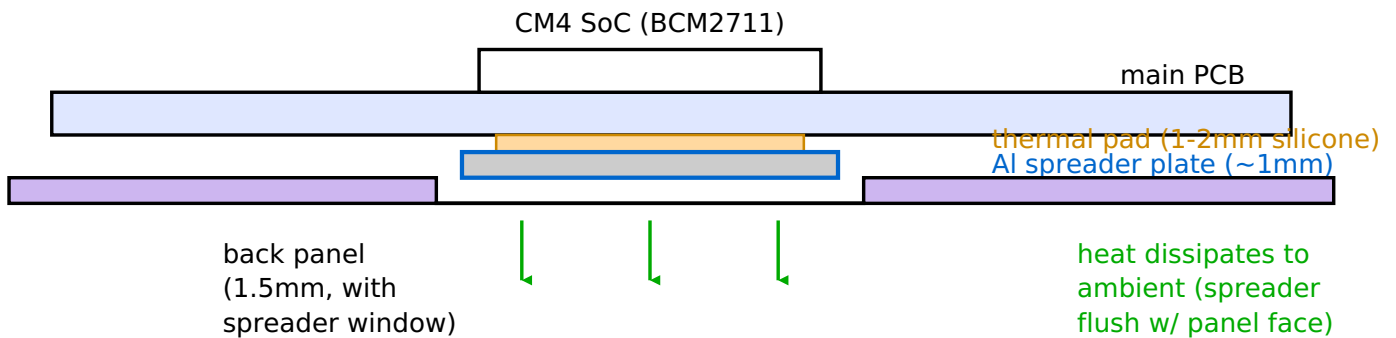
- **Sealed chamber** (not ported) — simpler to print/assemble for a prototype, and meaningfully improves bass response and efficiency vs. a speaker mounted with no enclosure at all. A vented/ported design could be explored later if bass response proves inadequate.
- **Size:** ~Ø18mm x 6mm deep (1mm clearance around the 16mm speaker) ≈ **1.5cm³** per chamber. This fits within the ~11.5mm of combined PCB+BATT+BACK depth available at the bottom corners, away from the keyboard mechanism (which occupies the front/CTRL-DISPLAY layer) and the battery footprint (60x90mm, assumed more central — confirm once battery placement is finalized).
- **Chamber walls:** printed integral to the back panel as a small recessed cup — speaker drops in from the front (PCB side) before the back panel is fastened on, sealing the chamber.
- **Grille:** 7x Ø2mm holes per speaker (~22mm² total open area) in a radial/flower pattern over the chamber — open area should be a meaningful fraction of the speaker cone area (~110-150mm² for a 16mm driver) without being so open that it weakens the enclosure or lets dust/debris in easily. 22mm² is a starting point; can increase hole count or diameter if sound output seems muffled in testing.
- **Orientation: rear-firing** (sound exits through the back panel), consistent with the BOM note "16mm speakers (rear/bottom-edge fire)". Bottom-edge-firing (grille on the bottom edge rather than back face) is an alternative if the back face needs to stay clean/flat — same chamber concept, grille relocated to the bottom edge wall instead of the back face.

Open follow-up: exact chamber X/Y placement depends on final battery placement and PCB component layout (both still TBD). The ~Ø18mm x 6mm chamber size and 7-hole grille pattern should transfer directly once placement is locked in — this page establishes the *design*, not the final coordinates.

back panel



B — CROSS-SECTION: SoC → thermal pad → spreader → ambient



Approach: passive only (no fan).

- **Heat source:** CM4's BCM2711 SoC is the dominant heat source — typical TDP ~3-5W under sustained Chromium/audio load (matches the ~500-700mA @5V CM4 estimate on the POWER BUDGET page).
- **Thermal path (B):** a 1-2mm silicone thermal pad sits between the SoC and a thin (~1mm) **aluminum spreader plate**, ~24x18mm. The back panel has a matching cutout/window so the spreader plate sits **flush with the outer back panel face** — turning part of the case itself into a heat-dissipating surface, rather than relying on resin (a poor conductor) to carry heat away.
- **Vent slots (A):** a single mid-height row (3x ~1.5 x 8mm) on the back panel, placed above the battery zone and clear of the speaker chambers. **Design decision:** the originally-planned upper row was dropped for a cleaner back face and less dust ingress — this sacrifices the through-flow chimney effect (single-row venting gives slow air exchange only), making the **spreader plate explicitly the primary cooling path**. If step-16 temperature measurements run hot, the documented fix is restoring an exit row — preferably as slots in the top edge wall rather than the back face (better chimney, back face stays clean). A reprint, not a redesign.
- **Placement:** CM4+CM4IO assumed in the upper-middle area of the device, away from the battery (lower 60x90mm) and the keyboard mechanism (front layer) — consistent with leaving the lower portion for battery + speaker chambers (per BATTERY and SPEAKER pages).

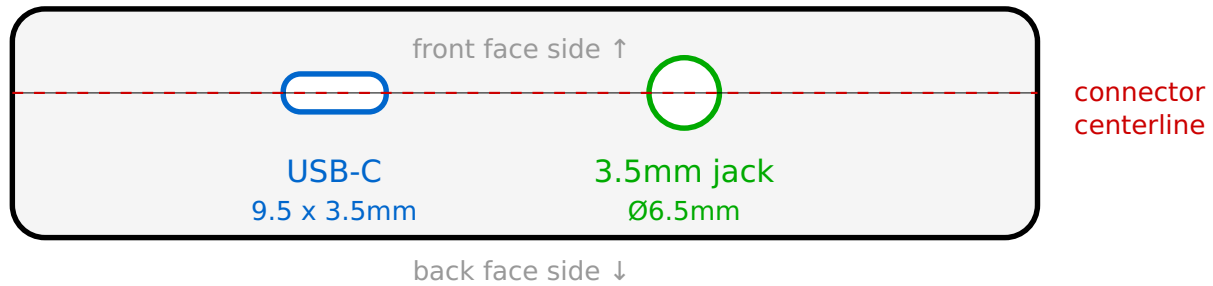
Sufficiency check: at ~3-5W, a flush aluminum spreader (~430mm² plate area) plus a few vent slots for passive convection is a well-trodden approach for similar Pi-based handhelds and should keep the SoC within safe operating temps for typical use. **This should be validated** with a thermal camera or simple temperature probe once the breadboard is running under load (build steps 15-16) — if temps run high, options include a larger spreader plate, more/larger vent slots, or (last resort) a small low-profile fan.

Open follow-up: exact CM4 placement and spreader window coordinates depend on final PCB layout — this page establishes the thermal *strategy* (flush spreader + passive vents), to be positioned precisely once the PCB floorplan is finalized.

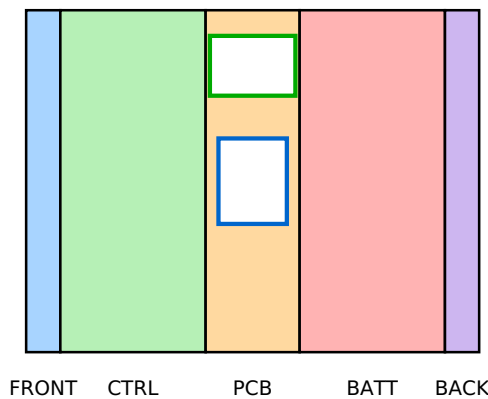
cy83rd3ck — USB-C & 3.5MM JACK PLACEMENT (CONCEPT)

A — BOTTOM EDGE VIEW (device standing, looking at the bottom edge)

Bottom edge — 94.6 x 21.1mm (speakers at corners, connectors centered between)



B — CROSS-SECTION AT TOP EDGE (connector alignment to PCB)



3.5mm: PJ-320A slim SMD, on PCB surface (~5mm tall) — barrel axis aligned to same edge

USB-C: 16-pin mid-mount receptacle — sits IN a PCB edge notch, connector centerline ≈ PCB centerline

Both connectors face the bottom edge wall; enclosure cutouts per view A.

Placement rationale:

- **Bottom edge** — USB-C and 3.5mm sit centered on the bottom edge, between the speaker chambers (which occupy the bottom corners, per the SPEAKER page). Bottom placement keeps the top edge clean (power button only) and puts the charge cable/headphone plug below the hands rather than in front of the screens.
- **USB-C at ~1/3 from left, 3.5mm at ~2/3** — spread apart so a charging cable and headphone plug don't crowd each other, both staying clear of the Ø18mm speaker chambers at the corners (~25mm of edge each side is speaker territory; the central ~45mm hosts both connectors comfortably).

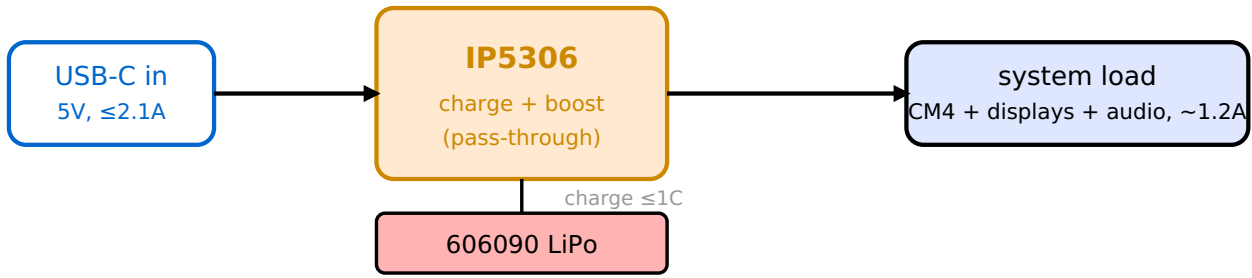
Vertical (depth) alignment:

- **USB-C:** 16-pin **mid-mount** receptacle (per BOM) sits in a notch cut into the PCB edge, putting the connector's centerline at the PCB's centerline — within the 4mm PCB layer, this lands the port roughly mid-depth of the device, leaving comfortable wall clearance on both sides. Enclosure cutout: **9.5 x 3.5mm** rounded slot.
- **3.5mm jack:** PJ-320A-class slim SMD (~5mm tall, per BOM) mounts on the PCB surface with its barrel facing the bottom edge. Enclosure cutout: **Ø6.5mm** round hole, centered on the barrel axis.
- Exact cutout Z-positions follow from the PCB's position in the depth stack (FRONT 1.5 + CTRL 6 + gap 0.5 puts the PCB layer 8-12mm from the front face) — final cutout coordinates to be taken from the PCB layout once routed, not hand-measured.

Open follow-up: exact X positions may shift with final PCB component routing (the IP5306 charge circuit wants to be near the USB-C; the WM8960 near the jack — both now live at the bottom of the board). Bottom-edge routing must coexist with the keyboard FSR FPC connector, also at the board's bottom — plan these side by side at PCB rev1. The edge choice, connector types, and cutout sizes here are stable.

cy83rd3ck — CHARGE-WHILE-PLAYING: THERMAL & SAFETY (CONCEPT)

POWER FLOW — charging while in use



The scenario: device plugged into USB-C while actively playing — the IP5306 must source the ~1.2A system load AND charge the battery simultaneously, all inside a sealed resin case.

Why it's manageable:

- The IP5306 is a power-bank SoC designed exactly for this (pass-through / charge-while-discharge is a standard power-bank behavior). Input is capped at ~2.1A: under simultaneous load it prioritizes the system, with remaining input budget going to battery charge — so worst-case input is bounded, not additive.
- Worst-case dissipation in the IP5306 itself: at ~85-90% conversion efficiency on ~10W throughput, expect ~1-1.5W of heat at the chip — significant for a small SOIC-8, but a known quantity.

Design mitigations (locked into the concept):

1. **Placement:** IP5306 sits near the USB-C connector at the **bottom edge** of the PCB (per the JACK PLACEMENT page). The back panel's mid-height vent slots (THERMAL MANAGEMENT page) sit above it — the nearest air exchange path for charge-circuit heat. The two heat sources stay physically separated (IP5306 bottom edge, CM4 mid/upper board).
2. **Copper pour:** at PCB layout time, give the IP5306 a generous ground/copper pour as a heatsink — standard practice, zero cost.
3. **Charge rate cap:** 606090 cell charged at ≤1C (≤4A) is far above what the IP5306 can deliver anyway (~2A charge max) — so the battery itself is never fast-charged beyond spec. Battery heat during charge is modest at this rate.
4. **Battery separation:** the battery sits in its own depth layer with the PCB between it and the hot components — it is not stacked against the IP5306 or CM4.

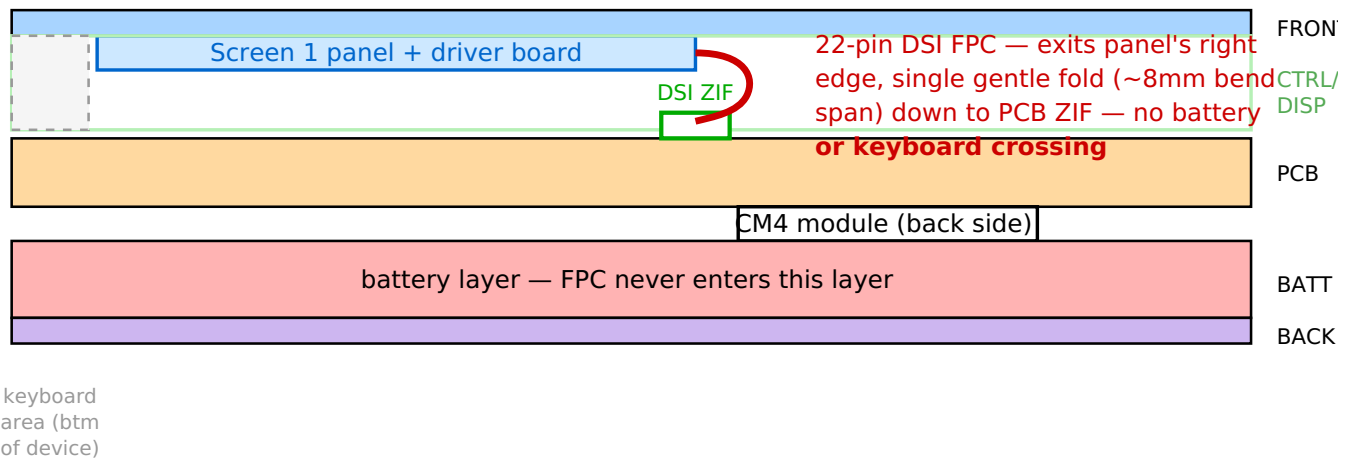
Safety notes:

- Use a 606090 cell **with built-in protection circuit (PCM)** — overcharge/overdischarge/short protection independent of the IP5306 (most off-the-shelf 606090s include this; confirm at purchase — the BOM line assumes it).
- The IP5306 itself provides overcharge voltage cutoff (4.2V), over-discharge cutoff, and short-circuit protection as built-in features — the PCM is a second, independent layer.

Validation (build steps 15-16): when measuring power draw and CM4 temps on the integrated breadboard, repeat the measurement **while charging** — probe the IP5306 region temperature specifically. If it runs hot in the enclosure, remedies are: larger copper pour, a small vent slot directly over the charge circuit, or limiting charge current (IP5306 boards typically allow this via the current-set resistor).

cy83rd3ck — SCREEN 1 DSI CABLE ROUTING (CONCEPT)

CROSS-SECTION — FPC path from panel to CM4, within the depth stack



Routing plan:

- The Waveshare 4-DSI-TOUCH-A's 22-pin DSI FPC exits the panel's driver board edge. The matching **22-pin ZIF socket sits on the front side of the main PCB, placed directly adjacent to the FPC's exit point** — so the cable makes one short, gentle fold (~8mm bend span) from the panel down to the PCB. Total cable run: ~15-25mm.
- **The FPC never crosses the keyboard mechanism or the battery layer.** The keyboard occupies the bottom of the device (panel's lower edge is well above it), and the battery sits behind the PCB — the FPC stays entirely within the CTRL/DISPLAY-to-PCB gap on the front side.
- **Bend radius:** standard 0.5mm-pitch FPC tolerates a ~2-3mm minimum bend radius; the available ~6mm of CTRL/DISPLAY-to-PCB depth gives a comfortable single S-fold with no creasing. Never crease an FPC flat — the gentle fold also acts as strain relief.
- **Slack/serviceability:** leave ~5mm of slack so the panel can be lifted out for service without disconnecting under tension; secure mid-run with a small foam pad or kapton tape dot to prevent rattle/vibration contact with the speaker-driven panel.
- **Touch FPC:** the panel's capacitive touch controller is on the same driver board and shares the DSI connector on this Waveshare model (I2C touch carried in the same harness via the DSI connector's I2C pins) — no separate second cable to route.

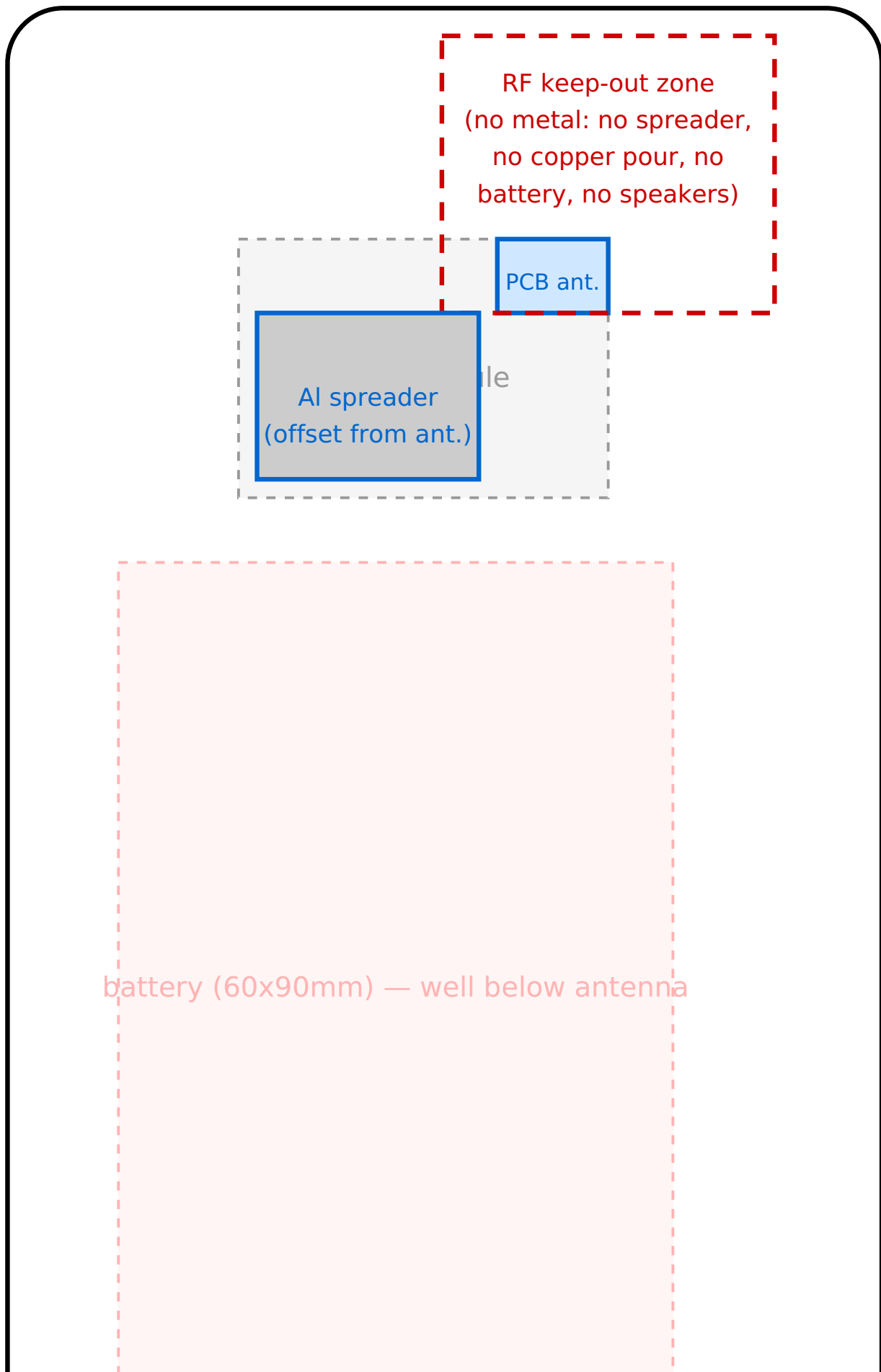
PCB layout note: this constrains the DSI ZIF's position on the main PCB — it must land adjacent to wherever the panel's FPC exit sits when the panel is in its mounted position (SCREEN 1 CUTOUT page). Lock this coordinate first when starting PCB rev1, since the panel's position is fixed by the front-panel cutout and can't move to accommodate the PCB.

Open follow-up: confirm the FPC exit point's exact location and orientation (which edge, facing which way) once the physical panel arrives — the routing concept holds either way, but the ZIF placement coordinate depends on it.

cy83rd3ck — WIFI/BT ANTENNA PLACEMENT (CONCEPT)

BACK VIEW — RF keep-out zone around the CM4 antenna corner

back panel view



Why this is mostly favorable already:

- **SLA resin is RF-transparent** — unlike a metal case, the enclosure itself barely attenuates 2.4/5GHz. The risk isn't the case; it's *metal objects near the antenna*: the battery pouch (aluminum laminate), the thermal spreader plate, speaker magnets, and large PCB copper pours.
- The CM4's onboard PCB antenna sits at **one corner of the module**. The fix is orientation: **rotate/place the CM4 so the antenna corner faces the top edge of the device**, pointing away from the battery (bottom 60x90mm), the speakers (bottom corners), and the keyboard FSRs (bottom).

Design rules (locked into the concept):

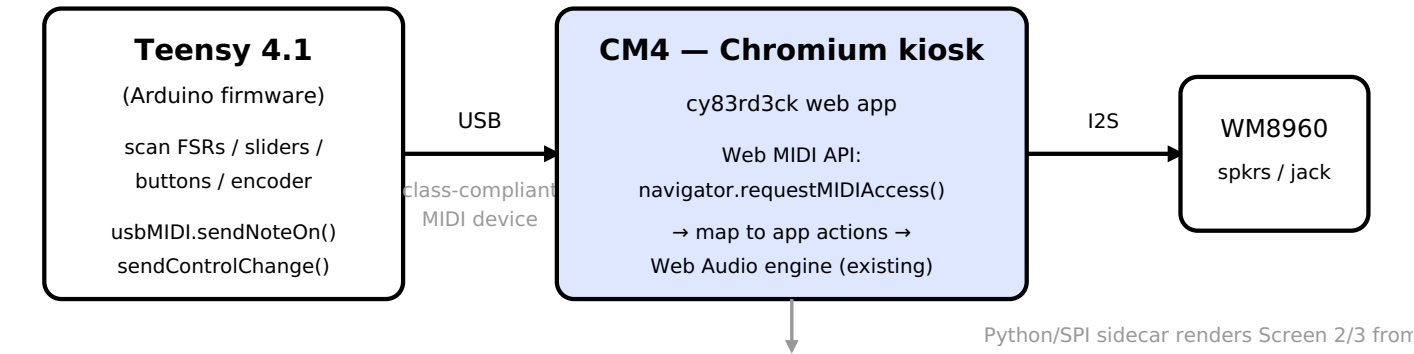
1. **CM4 orientation**: antenna corner toward the top edge. This is a free constraint to apply at PCB layout — it costs nothing if decided now, and is painful to fix after routing.
2. **RF keep-out zone**: ~15mm around the antenna corner — no copper pour on any PCB layer, no ground plane, no metal hardware (screws/standoffs), and critically: **the thermal spreader plate is offset away from this zone** (toward the device's left side, per the updated sketch — spreader and antenna must not overlap; both want the CM4's vicinity, so they take opposite sides of it).
3. **Battery clearance**: the 606090 sits in the lower 2/3 of the device, ~25mm+ from the antenna corner — fine.
4. **USB-C cable proximity**: no longer a factor — USB-C moved to the bottom edge, the full device length away from the antenna corner. The top edge hosts only the power button (plastic, no RF impact).

Fallback (if onboard antenna underperforms): the CM4 has a **U.FL connector** for an external antenna, selectable in software (dtparam=ant2). A thin adhesive flex antenna (~\$3-5) stuck to the inside of the resin back panel, away from all metal, is a well-proven fix used by many CM4 handheld builds — add the U.FL pigtail clearance to the back panel design only if testing shows it's needed.

Validation: during breadboard integration (build step 14), check WiFi RSSI/throughput with the CM4 positioned as planned, with the battery and spreader plate held in their approximate final positions — a 2-minute iwconfig/speed-test sanity check before committing the PCB layout.

cy83rd3ck — FIRMWARE / SOFTWARE INTEGRATION (CONCEPT)

ARCHITECTURE — input to sound, end to end



MIDI MESSAGE MAP (Teensy → web app)

Physical input	MIDI message	App action
Keyboard (12 keys)	Note On/Off, ch.1, notes 60-71 (C4-B4), velocity from FSR slope	Play note in active engine; octave nav buttons shift the note range in-app
BEND slider	Pitch Bend, ch.1	Pitch bend (existing app control)
MOD slider	CC1 (mod wheel), ch.1	Mod (existing app control)
KICK/SNRE/HiHAT/BASS/LEAD	CC20-24, value 127 on press	Track select
TAP / NOTE / RPT / LP	CC25-28	Transport/mode toggles
Tab row (10)	CC30-39	Switch active tab (Seq/Engine/.../H4CK)
Octave nav + sound select (12)	CC40-51	Octave shift, sound/sample select
RECORD encoder turn	CC52, relative (1=CW, 127=CCW)	Parameter scroll / value adjust
RECORD encoder push / PLAY	CC53 / CC54	Record arm / play-stop

Why this architecture is low-risk:

- **Teensy's USB MIDI is class-compliant** — the CM4 needs no driver; the Teensy enumerates as a standard MIDI device the moment it's plugged in (Arduino IDE → USB Type: MIDI; usbMIDI.sendNoteOn(...) etc. are built into the Teensy core).
- **Chromium supports the Web MIDI API natively** — navigator.requestMIDIAccess() in the existing cy83rd3ck app receives the messages directly. No middleware, no native code on the CM4 for the input path. The kiosk launch flag --autoplay-policy=no-user-gesture-required (and granting MIDI permission for the app's origin) makes it hands-free at boot.
- **The web app's sound engine already exists** — the integration work is a single new input module: a Web MIDI listener that maps the message table above onto the same app actions the on-screen UI already triggers. The physical controls become an alternate input path, not a rewrite.

The genuinely new piece: the **Screen 2/3 sidecar** — a small Python service on the CM4 that renders the OLEDs over SPI from app state. Simplest bridge: the web app exposes state via a tiny WebSocket (or localStorage polled by a local endpoint), and the Python service draws waveform/status using the luma.oled library. This is the largest unknown in the software stack — prototype it early (can be developed today with just a CM4/any Pi + one OLED, no other hardware needed).

OLED burn-in protection (build into the sidecar from day one — risk #21): mono OLEDs have no built-in wear compensation, and static status layouts are the worst case for permanent ghosting. Four mitigations, all software: **(1) pixel shifting** — orbit the whole frame 1-2px every few minutes (the highest-value fix); **(2) dim by default** — run at ~50-60% contrast, wear scales steeply with brightness; **(3) idle screen-off** — blank Screens 2/3 after a few minutes of no MIDI/UI activity, wake instantly on input; **(4) layout habit** — avoid never-moving 100%-on regions (inverted-text header bars are the worst offender).

Firmware scope (Teensy side): one Arduino sketch — mux scan loop (FSR RESPONSE page), slider/button/encoder reads, debounce, and MIDI sends. All well-trodden Teensy territory; the only tuning-sensitive part is the FSR velocity constants, already covered on the FSR RESPONSE page.

Open follow-up: CC number assignments above are placeholders (any free CC works) — finalize when writing the sketch + app listener together. Add a MIDI map constants file shared in spirit between the two codebases so they can't drift.

cy83rd3ck — ENCLOSURE CLOSURE METHOD (CONCEPT)

A — SCREW BOSS LAYOUT (back view, 6x M2)

back panel — screws enter from the back



M2



M2

bosses hug the perimeter wall —
clear of battery, speakers, vents,
spreader window & PCB footprint



M2



M2

battery

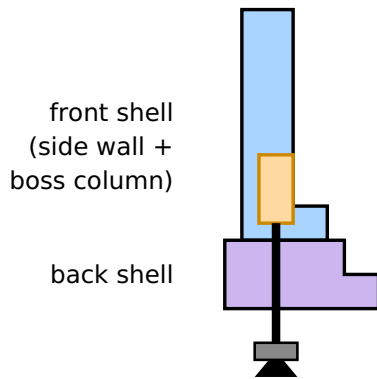


M2



M2

B — CROSS-SECTION AT A BOSS (lip-and-groove seam + heat-set insert)



lip-and-groove seam:

0.8mm lip on front shell wall nests into matching groove on back shell — aligns the shells, hides the seam line, blocks dust

M2 heat-set brass insert
(melted into resin boss)

M2x6 screw, countersunk/recessed
head in back shell (enters from back)

Decision: M2 screws + heat-set brass inserts, entering from the back. 6x positions (4 corners + 2 mid-edge — top-right and bottom-right shifted per the BACK FACE COMPOSITE page to clear the RF keep-out and speaker chamber).

Why screws over snap-fit or adhesive:

- **Serviceability is the whole point of a prototype** — this device will be opened dozens of times during bring-up (battery, FSR strip tuning, PCB swaps). Snap-fits in SLA resin are the wrong material choice anyway (resin's low fatigue tolerance = cantilever snaps break after a handful of cycles — same reason the living hinge is flagged as risk #2). Adhesive is one-way.
- **Heat-set brass inserts** rather than threading into bare resin: resin threads strip after a few open/close cycles; brass inserts (M2 x 3mm, ~\$5/50pc) melt in with a soldering iron and survive hundreds of cycles. One-time tool cost: an insert tip for the soldering iron (~\$10).

Geometry:

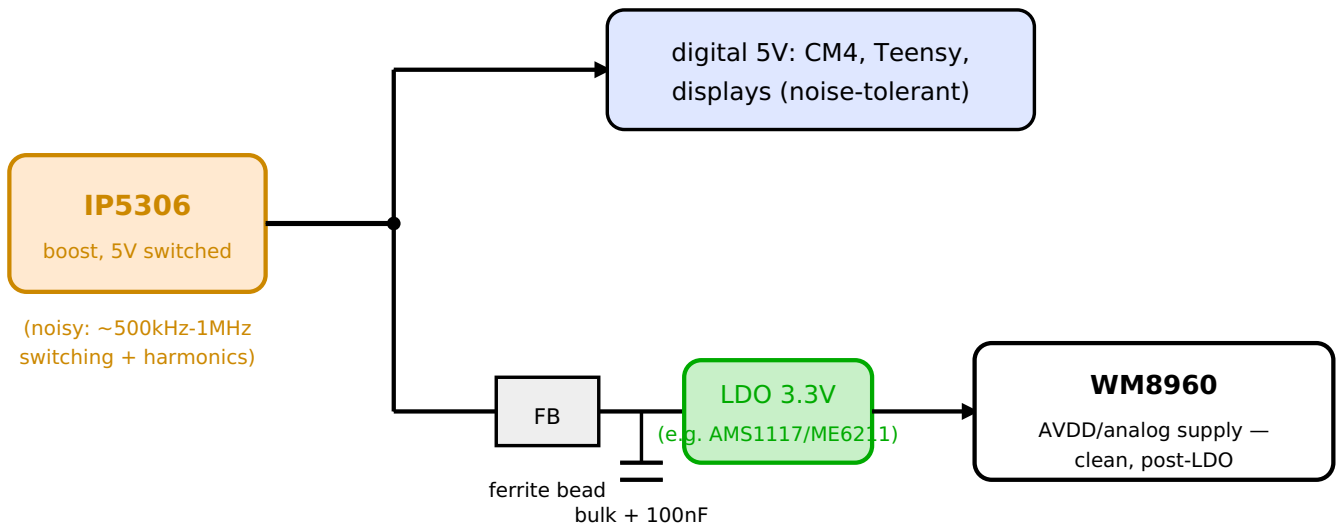
- **Boss columns are part of the front shell's perimeter wall** (thickened wall sections, ~Ø5mm), not freestanding posts — stronger in resin, and they consume zero interior floor space beyond the wall line. The 6 positions hug the perimeter, clear of the battery (interior), speakers (bottom corners are mid-edge-adjacent, bosses sit at the very corners), vent slots, and the spreader window.
- **Screws enter from the back** — the front face stays clean (no visible fasteners around screens/keys); recessed screw heads in the back shell sit flush.
- **Lip-and-groove seam:** a 0.8mm lip runs around the front shell's wall edge, nesting into a matching groove in the back shell — self-aligning during assembly, hides the seam line cosmetically, and keeps dust out without a gasket.
- **Stack impact:** bosses live within the existing wall/TOL allowances — no change to the 21.1mm depth or 94.6 x 164.46mm footprint. The screw length (M2x6) engages ~3mm of insert from a 1.5mm back panel + 1.5mm of boss engagement.

BOM additions: 6x M2x6 screws, 6x M2x3 heat-set inserts (~\$2 total at qty); insert installation tip (~\$10, one-time tool).

Open follow-up: exact boss X/Y coordinates get locked when the front-panel SCAD gains its side walls (currently it's the flat plate only) — add bosses, lip, and back shell as the next CAD work item (build step 17).

cy83rd3ck — AUDIO POWER NOISE ISOLATION (CONCEPT)

POWER TREE — keeping switching noise out of the audio path



The problem: the IP5306's boost converter switches at high frequency, and its noise (fundamental + harmonics, plus load-transient ripple from the CM4) rides on the 5V rail. If the WM8960's analog supply sees that ripple, it shows up as audible whine/hiss/"digital hash" in the speakers and headphone out — the classic flaw of cheap Pi audio builds.

The fix (three layers, all standard, ~\$1 total):

1. **Rail split:** digital loads (CM4, Teensy, displays) take the raw 5V — they don't care. The audio branch gets its own spur off the rail.
2. **Ferrite bead + bulk capacitance** on the audio spur: blocks the high-frequency switching content before it reaches the audio circuitry (bead sized for $\sim 600\Omega$ @100MHz, plus $\sim 100\text{-}220\mu\text{F}$ bulk + 100nF ceramic).
3. **Dedicated LDO for the WM8960's analog supply** (3.3V, e.g. ME6211/AMS1117-class): a linear regulator's PSRR strips the remaining ripple — the codec's analog section never sees the switcher. The WM8960's digital I/O can share the CM4's 3.3V domain as required for logic levels.

Layout rules for PCB rev1 (as important as the schematic):

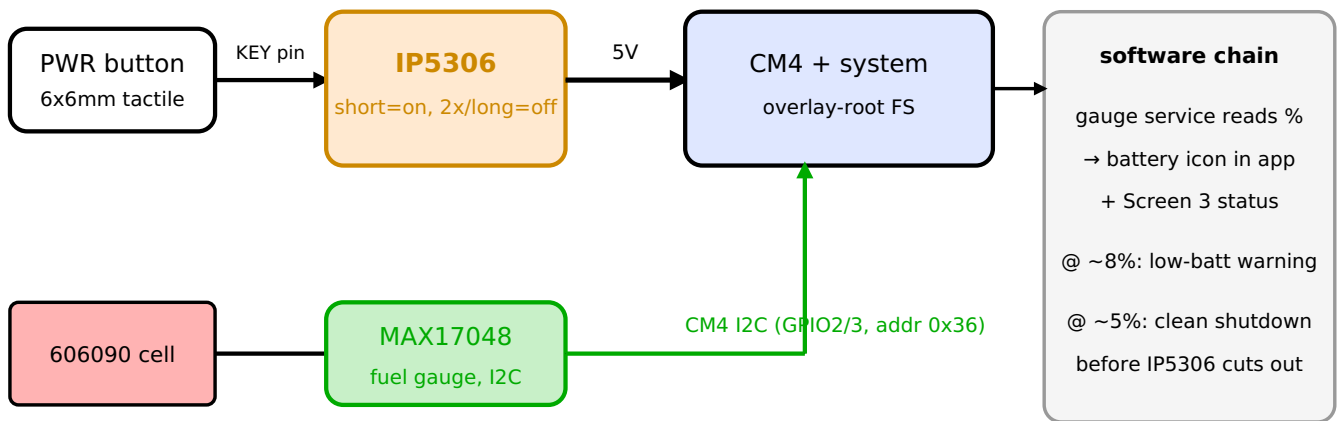
- **Physical separation:** IP5306 + its inductor live at the bottom edge near USB-C (per JACK PLACEMENT page); WM8960 + jack also want the bottom edge — keep them at **opposite ends of the bottom edge** (IP5306 by USB-C at $\sim 1/3$, WM8960 by the jack at $\sim 2/3$), with the boost inductor's loop area minimized and pointed away from audio traces and the FSR FPC connector.
- **Ground strategy:** single ground plane, but audio components grouped so their return currents don't share paths with the boost converter's switching loop (star-style return for the analog section; do not route audio traces over the switcher's ground return).
- **Speaker wiring:** speaker lines are amplified outputs (robust), but keep them away from the FSR analog lines (high-impedance, susceptible) — they run to opposite areas of the board anyway (speakers bottom, FSR connector bottom — route on opposite sides/layers).

Validation (build step 8): when audio first runs on the breadboard, listen at idle and under CM4 load (scroll the UI, WiFi transfer) with headphones at full volume — switching whine is unmistakable. Breadboard wiring is noisier than any PCB, so if it's quiet on the breadboard with the filter parts in place, the PCB will be quieter still. If whine persists on the PCB: increase bulk capacitance, upgrade the LDO's PSRR, or add a second bead on the WM8960's AVDD pin specifically.

BOM additions: 1x ferrite bead, 1x 3.3V LDO, $\sim 4\text{x}$ caps — $\sim \$1$ total, folded into the existing "misc" line.

cy83rd3ck — POWER BUTTON, BATTERY GAUGE & SAFE SHUTDOWN (CONCEPT)

SIGNAL FLOW — power control & battery awareness



#27 — Power button:

- A **6x6mm tactile on the top edge** (sole occupant of that edge — USB-C/jack live on the bottom edge). The KEY-pin trace runs the board length down to the IP5306 at the bottom edge — fine, it's a low-speed button signal. Top placement avoids accidental presses while playing and keeps power control away from the cable area.
- Behavior comes free from the IP5306: **short-press = on, double-press or long-press = output off**. No firmware needed for basic on/off.
- Enclosure: one more small cutout on the top edge (Ø4mm round or 4x4mm square, matching the button cap) — add to the SCAD when the side walls are modeled (build step 17).

#29 — Battery gauge: MAX17048 fuel gauge (~\$1, I2C addr 0x36) on the CM4's existing I2C bus (GPIO2/3, shared with the WM8960 — different addresses, no conflict; this is a normal multi-device I2C bus, not a two-master situation).

- Chosen over the alternatives: the I2C-variant IP5306 has spotty documentation/availability; a bare ADC divider gives voltage only (LiPo voltage→charge% is nonlinear and load-dependent — the MAX17048 does proper modeling for \$1).
- A small Python service (folded into the Screen 2/3 sidecar) polls the gauge and feeds: a battery icon/percent in the web app (via the existing WebSocket bridge) and the Screen 3 status display.

#28 — Safe shutdown, two independent layers:

- **Layer 1 — overlay root filesystem** on the CM4 (RasPi OS's built-in overlay FS mode): the OS partition mounts read-only with changes held in RAM, so **any** hard power cut is harmless to the system. Pairs naturally with the risk-#23 tmpfs config. App data (IndexedDB) lives on a small separate writable partition — flushed quickly after saves, tiny corruption window.
- **Layer 2 — graceful low-battery shutdown**: the gauge service issues a low-battery warning at ~8% (app banner + Screen 3) and a clean shutdown -h now at ~5%, before the cell sags to the IP5306's hard cutoff. User power-off via the button is a hard cut — harmless because of Layer 1.
- *Accepted gap*: unsaved in-app work is lost on button power-off (like any hard power switch). If that stings later, the app can autosave to IndexedDB on a short debounce — software-only fix, no hardware hook needed.

BOM additions: 1x 6x6mm tactile (power), 1x MAX17048 (~\$1) — folded into misc/logic lines. **PCB rev1 items:** KEY-pin button placement, MAX17048 on CM4 I2C. **Software items:** overlay FS + gauge polling in the sidecar (build step 1 and the integration page's sidecar work).

cy83rd3ck — SHOPPING LIST (REMAINING PURCHASES)

Already purchased: CM4108016, CM4IO Board, Teensy 4.1, Screen 1 (Waveshare 34354), 1x FSR, breadboard kit, multimeter. Below is everything still to order, grouped by when it's needed.

NEEDED FOR BREADBOARD PHASE (build steps 3-14)

☐	Item	Qty	Est.	Notes
☐	Screen 2 — 3.12" SSD1322 OLED	1	~\$15	SPI
☐	Screen 3 — 2.3" SSD1306 OLED	1	~\$8	SPI
☐	ALPS RS30111 slide pots	2	~\$8	BEND/MOD
☐	WM8960 codec breakout	1	~\$10	I2S+I2C
☐	BeStar 16mm speakers	2	~\$4	
☐	Interlink FSR 400 (7.62mm)	11	~\$50	FSR 400 preferred over 402 — see risk #18; confirm nub contact at test print before bulk order (order 1-2 first)
☐	CD74HC4067 mux breakout	1	~\$2.50	
☐	MCP23017	2	~\$3	I2C GPIO expanders
☐	MAX17048 fuel gauge breakout	1	~\$2	battery %
☐	Tactile switches: 12x12mm (x6), 6x6mm (x17), 4.5x4.5mm low-profile (x10)	set	~\$8	incl. power button
☐	EC11 rotary encoder w/ push	1	~\$2	RECORD
☐	606090 LiPo ~4000mAh with PCM	1	~\$8-12	confirm protection circuit included
☐	IP5306 charge/boost breakout	1	~\$4	
☐	SLA resin (tough/durable)	1 bottle	~\$15	test coupons + comb + panels

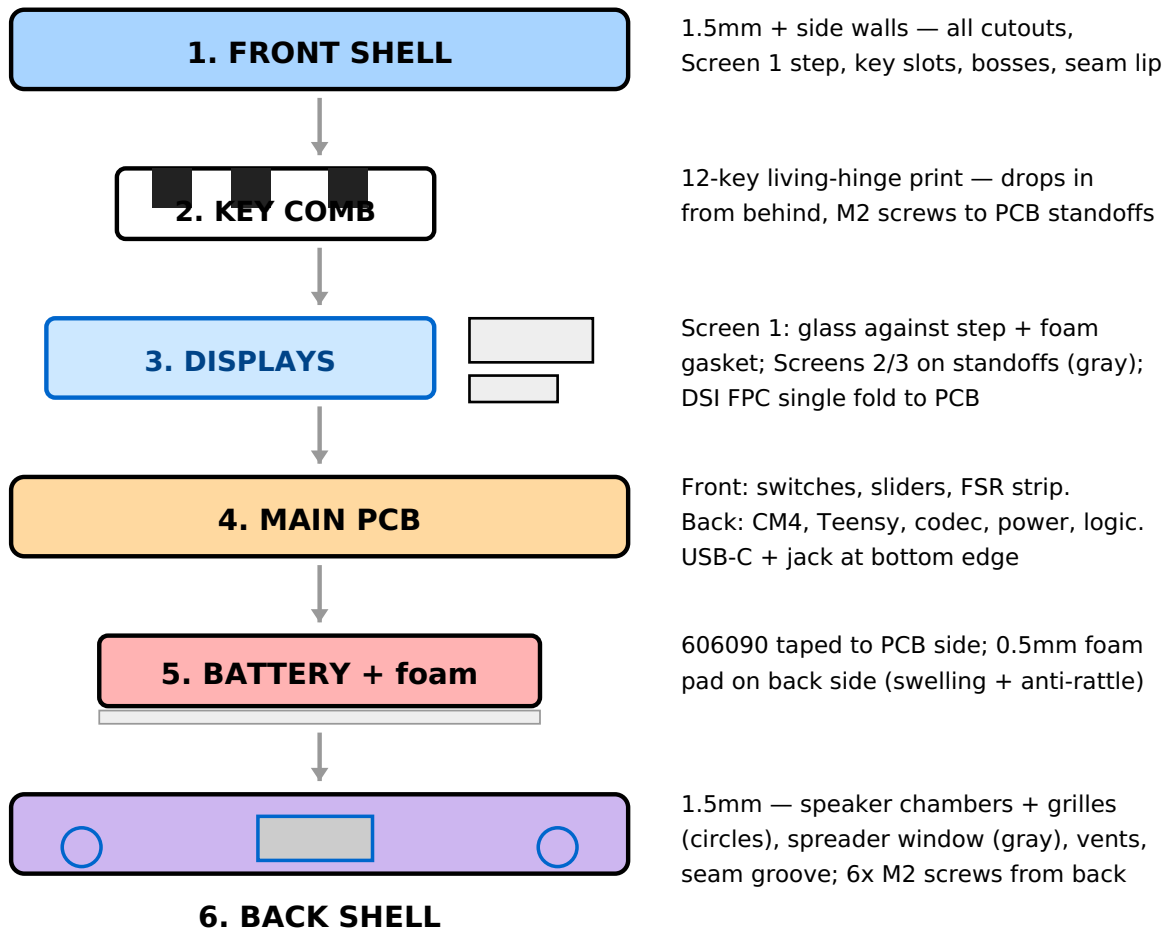
NEEDED FOR PCB REV1 / FINAL ASSEMBLY (steps 17-20)

☐	Item	Qty	Est.	Notes
☐	TVS arrays (TPD4E05 or similar)	3	~\$0.50	FSR ESD, PCB rev1
☐	Ferrite bead + 3.3V LDO + caps	set	~\$1	audio supply filter, PCB rev1
☐	Aluminum plate ~24x18x1mm + thermal pad	1	~\$3	spreader
☐	M2x6 screws + M2x3 heat-set inserts	6+6	~\$2	closure
☐	Heat-set insert tip (soldering iron)	1	~\$10	one-time tool
☐	Adhesive foam sheet ~0.5mm	1	~\$3-5	battery pad, gaskets
☐	14-pin 0.5mm FPC + ZIF sockets	1+1	~\$3	FSR strip link
☐	USB-C 16-pin mid-mount + PJ-320A jack	1+1	~\$2	PCB rev1
☐	Hirose DF40C-100DS-0.4V (CM4 mating receptacles)	2	~\$6	PCB rev1 — footprint from CM4IO reference

Prices are rough market estimates — check the COST SUMMARY page for totals, and update both pages as real receipts come in.

cy83rd3ck — EXPLODED ASSEMBLY VIEW (CONCEPT)

The six layers, separated — top of diagram = front of device (what you touch), bottom = back of device

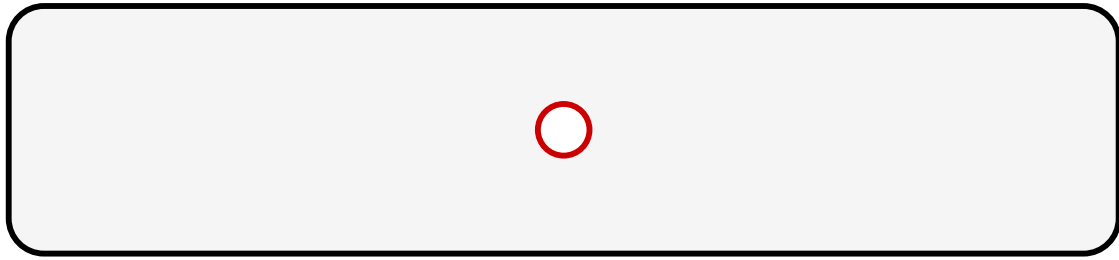


Assembly order (reverse of the explosion, i.e. bottom-up onto the PCB):

1. Populate the PCB (both sides).
 2. Mount displays and the key comb onto the PCB standoffs.
 3. Tape the battery to the PCB's back; stick the foam pad on the battery's back.
 4. Lower this whole assembly into the front shell — Screen 1's glass seats against its step, keys pass through their slots.
 5. Drop the speakers into their back-shell chambers, fit the back shell (lip into groove), drive the 6x M2 screws.
 6. Fit the spreader plate + thermal pad in the back-panel window.
- Everything is hand-assembled and fully reopenable.

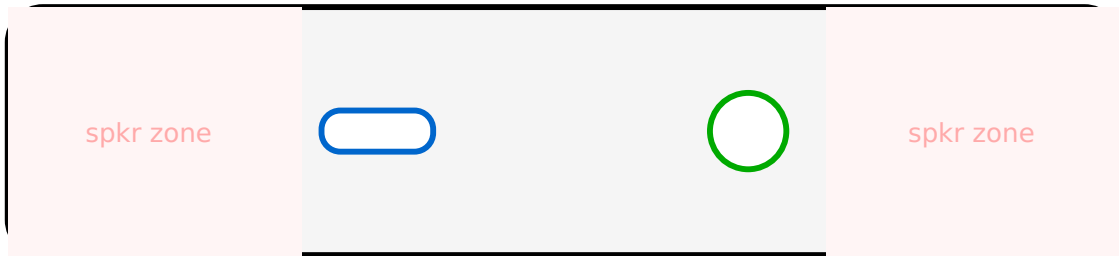
cy83rd3ck — EDGE ELEVATIONS (ALL FOUR EDGES)

TOP EDGE (94.6 x 21.1mm) — power button only



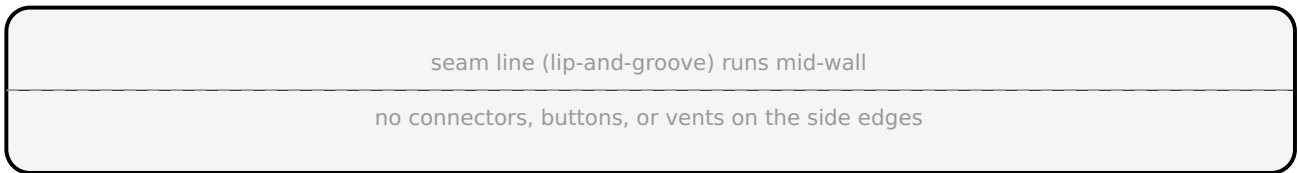
PWR button, Ø4mm, centered

BOTTOM EDGE (94.6 x 21.1mm) — USB-C + 3.5mm between speaker corners



USB-C 9.5x3.5 @ ~1/3 3.5mm Ø6.5 @ ~2/3

LEFT & RIGHT EDGES (164.46 x 21.1mm) — clean, no cutouts



Back face (not an edge, for completeness): vent slots (single mid row), spreader plate window, speaker grilles (2x 7-hole patterns at bottom corners), 6x recessed M2 screw heads — all per the THERMAL, SPEAKER, and CLOSURE pages.

Front face: screens, controls, and keys only — zero fasteners or I/O.

Status: these elevations resolve the risk #33 check; exact coordinates transfer to the SCAD when side walls are modeled (build step 17).

cy83rd3ck — NET LIST (FOR KiCad SCHEMATIC ENTRY)

Every electrical connection in the design, exhaustive enough that entering the schematic into KiCad is transcription, not design. Start the KiCad project from the official CM4IO reference files (already sourced). Net names in CAPS. Teensy direct-GPIO pin numbers marked * are placeholder assignments — any free pin works, finalize in the sketch.

POWER

Net	From	To	Notes
VBUS_IN	USB-C VBUS (via CC 5.1k pulldowns x2)	IP5306 VIN	5.1kΩ on CC1/CC2 for 5V/3A advertisement
BAT+	606090 cell + (via PCM)	IP5306 BAT, MAX17048 CELL	gauge senses cell directly
5V_SYS	IP5306 OUT	CM4IO 5V in, Teensy VIN, Screen1 5V, audio filter input	main rail
5V_AUD	5V_SYS via ferrite bead FB1 + 220μF + 100nF	LDO U_AUD VIN	audio spur, see AUDIO NOISE page
3V3_AUD	LDO U_AUD (ME6211 3.3V) OUT + 10μF	WM8960 AVDD	clean analog supply
3V3_CM4	CM4IO 3.3V out	WM8960 DVDD/logic, Screen2 VCC, Screen3 VCC, MAX17048 (if not cell-powered variant)	
3V3_T	Teensy 3.3V out	CD74HC4067 VCC, MCP23017 x2 VDD, FSR strip 3.3V rail, sliders, encoder/button pullups	≤250mA budget — these loads are tiny
PWR_KEY	PWR tactile (top edge), other side GND	IP5306 KEY	short=on, double/long=off
GND	common: all of the above + both shells' hardware		single plane, audio returns grouped (AUDIO NOISE page)

CM4 (via CM4-style socket / CM4IO reference)

Net	CM4 pin	To	Notes
(mating)	CM4 200-pin interface	2x Hirose DF40C-100DS-0.4V receptacles on PCB back	standard CM4 board-to-board system; copy footprint/pinout from CM4IO reference design
DSI0_*	DSI connector (22-pin)	Screen 1 FPC (incl. touch I2C in-harness)	ZIF adjacent to panel exit, DSI ROUTING page
I2C1_SDA / I2C1_SCL	GPIO2 / GPIO3	WM8960 (0x1A), MAX17048 (0x36)	2.2-4.7kΩ pullups to 3V3_CM4
I2S_*	GPIO18/19/20/21	WM8960 BCLK/LRCLK/ADCDAT/DACDAT	
SPI0_MOSI/SCLK	GPIO10 / GPIO11	Screen2 + Screen3 (shared)	
SPI0_CE0 / CE1	GPIO8 / GPIO7	Screen2 CS / Screen3 CS	
OLED_DC / OLED_RST	GPIO25 / GPIO24	Screen2+3 D/C, RST (shared)	any free GPIO
USB_HOST	CM4 USB D+/D-	Teensy 4.1 USB device port	MIDI/serial
ANT	onboard / U.FL	—	antenna corner to top edge, RF keep-out (WIFI page)

TEENSY 4.1 (input controller)

Net	Teensy pin	To	Notes
FSR_SIG	A0 (14)	CD74HC4067 SIG	
MUX_S0-S3	2,3,4,5 *	CD74HC4067 S0-S3	EN→GND
BEND / MOD	A1 (15) / A2 (16)	RS30111 #1 / #2 wipers	pot ends: 3V3_T and GND
BTN_TAP/NOTE/RPT/LP	6,7,8,9 *	4x 6x6mm tactiles → GND	INPUT_PULLUP, no external R
BTN_TRACK1-5	10,11,12,28,29 *	5x 12x12mm tactiles → GND	INPUT_PULLUP
BTN_PLAY	30 *	PLAY tactile → GND	INPUT_PULLUP
ENC_A/B/SW	31,32,33 *	EC11 A/B/push → GND	INPUT_PULLUP
I2C2_T (Wire2)	25 (SDA2) / 24 (SCL2)	MCP23017 #1 (0x20) + #2 (0x21)	2.2-4.7kΩ pullups to 3V3_T. Wire2, not Wire1 — pins 16/17 are reserved for analog (A2/A3); see conflict note below.
USB_DEV	micro-USB	CM4 USB host	

FSR STRIP & EXPANDERS

Net	From	To	Notes
FSR_C0-C11	each FSR low side (high side → 3V3_T)	10kΩ→GND at node; node → TVS (TPD4E05 x3) → mux C0-C11	via 14-pin FPC: 12 sig + 3V3_T + GND
TAB1-10	MCP23017 #1 GPA0-7, GPB0-1	10x 4.5mm tactiles → GND	internal pullups
OCT/SND1-12	#1 GPB2-7 + #2 GPA0-5	12x 6x6mm tactiles → GND	internal pullups; #2 GPA6-7, GPB0-7 spare

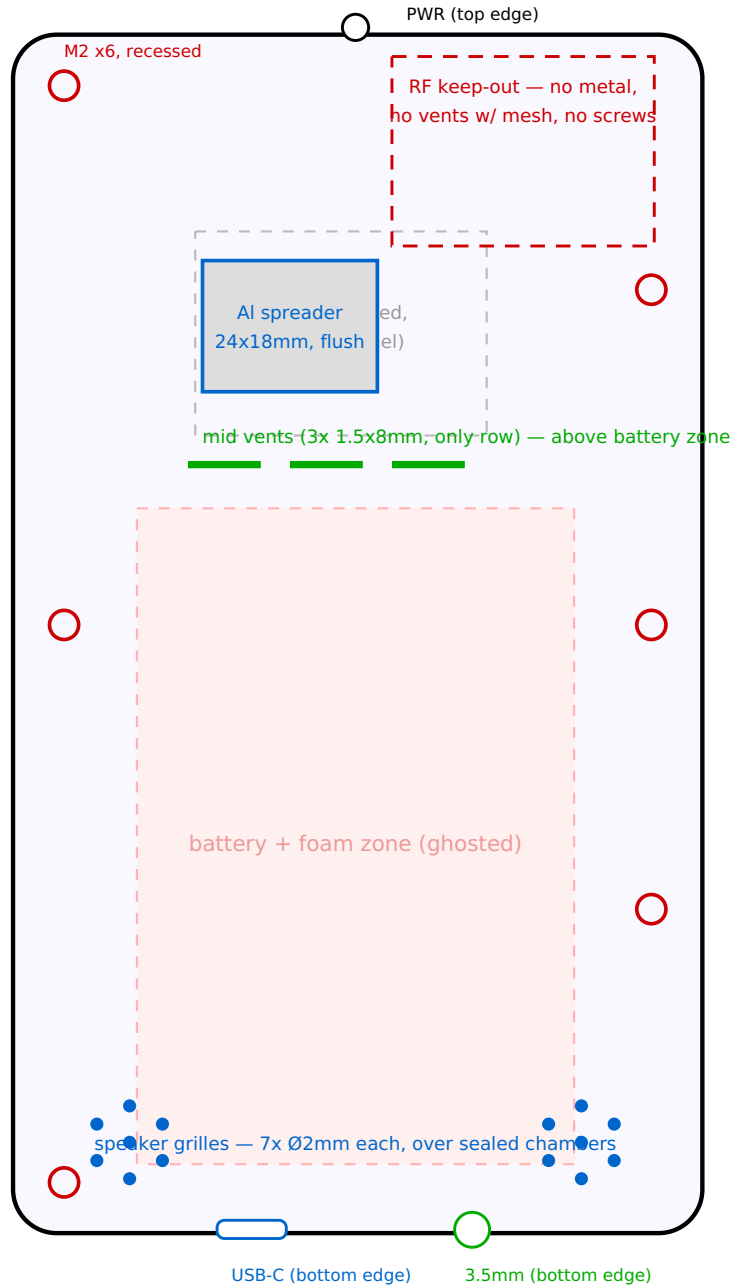
AUDIO OUT

Net	From	To	Notes
SPK_L± / SPK_R±	WM8960 SPK_LP/LN, SPK_RP/RN	16mm speakers L/R	class-D bridge out, keep pairs together, away from FSR FPC
HP_L/R/GND	WM8960 HP_L, HP_R, OUT3/HPGND	PJ-320A jack tip/ring/sleeve	jack detect optional → WM8960 JD

One real conflict caught while compiling this: Teensy pins 16/17 serve double duty (A2/A3 analog vs Wire1 SDA1/SCL1). Resolution recorded above: keep BEND/MOD on A1/A2 (pins 15/16) is NOT possible alongside Wire1 — so the MCPs move to **Wire2 (pins 24/25)**, and the PIN ASSIGNMENT page should be read with that correction. Exactly the kind of error a net list exists to catch before wiring.

cy83rd3ck — BACK FACE COMPOSITE (ALL FEATURES, ONE VIEW)

Back panel, 94.6 x 164.46mm — every feature from the THERMAL, SPEAKER, CLOSURE & ANTENNA pages, drawn together



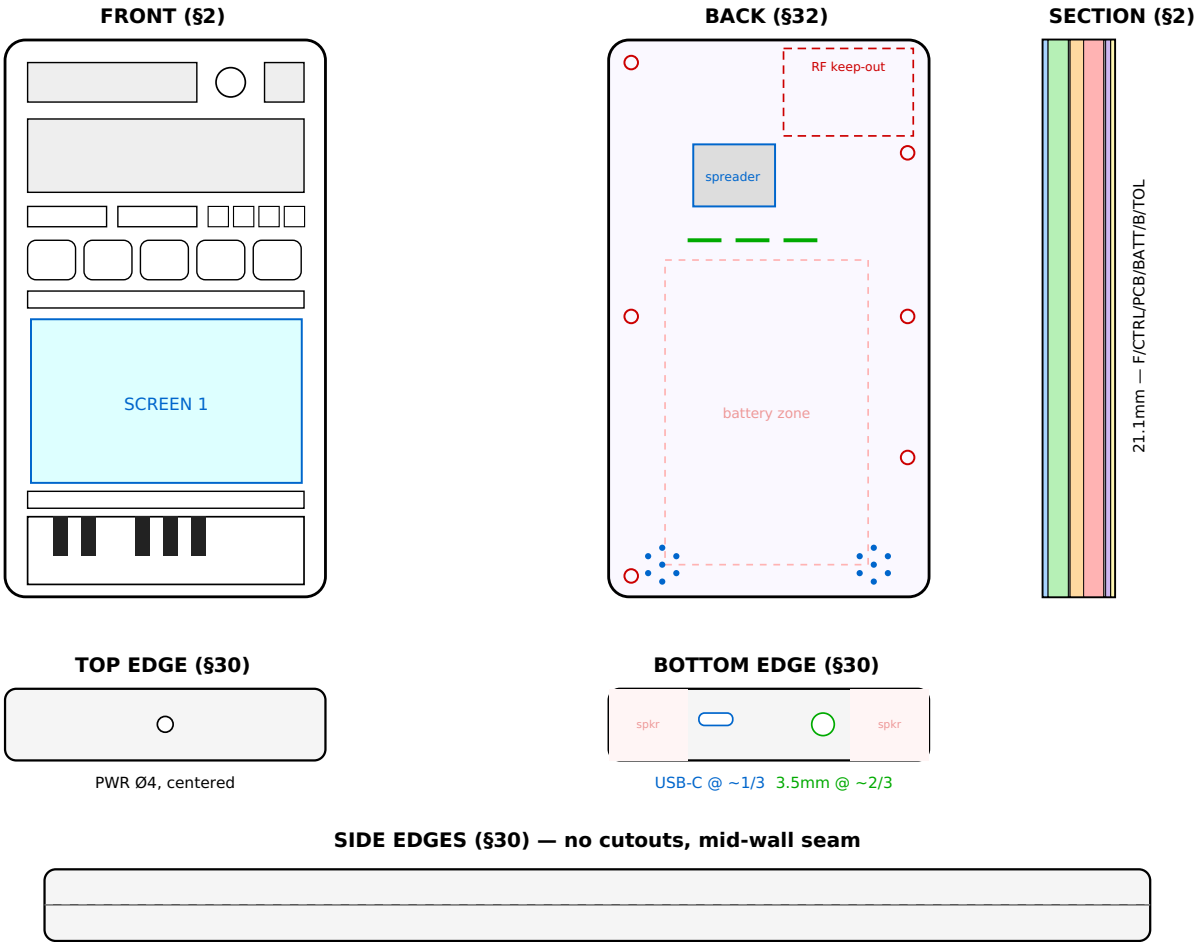
Collision checks performed by drawing this composite:

- **Screw positions vs everything**: the 6 perimeter bosses moved where needed — top-right screw shifted down out of the RF keep-out (no metal in the zone), bottom-right screw moved up clear of the right speaker chamber. Final set: 4 corners (with those two adjustments) + 2 mid-edge. Update the CLOSURE page coordinates to match at step 17 CAD.
- **Mid vents vs battery foam**: mid vent row sits just above the battery zone's top edge — vents breathe into the gap above the battery, not into the foam pad. Clear.
- **Spreader window vs keep-out**: spreader sits left of the CM4, outside the antenna zone. Clear.
- **Grilles vs screws/bosses**: bottom-corner screws sit outboard of the grille patterns. Clear after the bottom-right adjustment.

Drawing set status: with this page, the enclosure now has a complete set — front face (§2), back face (this page), all four edges (§30), depth cross-section (§2), and exploded stack (§29). Everything needed to model the full case in CAD at build step 17.

cy83rd3ck — FULL DRAWING SET (ONE SHEET, ALL VIEWS)

Orthographic set, simplified — for detail dimensions see the referenced sections. Device: 94.6 x 164.46 x 21.1mm



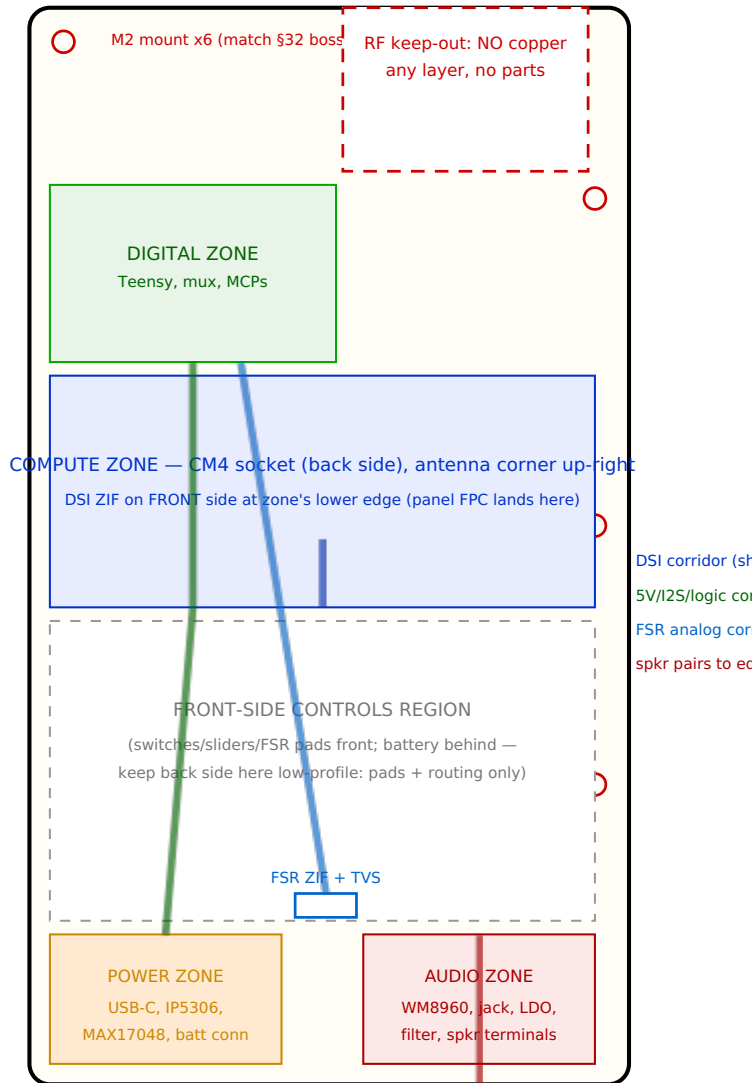
164.46mm — lip-and-groove seam line shown dashed; left and right edges identical

Views simplified and not to a common scale — authoritative dimensions live in the referenced sections (§2, §17-18, §20, §25, §30, §32).

cy83rd3ck — PCB LAYOUT PLAN (PRE-ROUTING)

The bridge between the floorplan (§4) and KiCad (step 18): board outline, stackup, zones, and routing corridors

board outline ~88 x 158mm (case interior minus walls/clearance)



Stackup decision: 4-layer (sig / GND / 3V3-5V split power / sig), standard 1.6mm. A 2-layer board is conceivable but fights you everywhere: DSI wants a continuous reference plane, the audio section wants clean ground separation from the boost converter, and the FSR analog lines want shielding from the speaker pairs. At prototype quantities the 4-layer premium is a few dollars per board — take it.

Zone logic (matches §4 floorplan + later decisions): power and audio occupy opposite ends of the bottom edge (§20, §26); the FSR ZIF + TVS arrays sit between them at the board's bottom (§15, risk #24); compute mid-board with the antenna corner into the keep-out (§23); digital I/O upper-left; the whole front-side controls region keeps its back side low-profile because the battery presses against it (§29 stack).

Routing corridor rules:

- **DSI:** shortest corridor on the board (ZIF lands adjacent to the panel FPC, §22) — length-matched differential pairs over solid ground, no layer changes if possible.
- **FSR analog corridor** (12 lines, bottom ZIF → mux in digital zone): route on the side/layer opposite the speaker pairs; these are the board's most noise-sensitive nets (high-impedance) next to its noisiest neighbors.
- **I2S corridor** (CM4 → audio zone): ordinary digital, but keep it out of the audio zone's analog half — enter from the WM8960's digital side.
- **Boost inductor loop** (power zone): minimize loop area, orient away from the audio zone; no sensitive routing underneath on any layer.
- **KEY trace** (top-edge button → IP5306): full board length, any path — it's a button; route it last through whatever's left.

Hard coordinates to lock first in KiCad (in this order): 1) board outline + 6 mounting holes from the enclosure CAD (step 17 feeds step 18), 2) DSI ZIF position from the mounted panel's FPC exit (confirm when panel arrives — §22), 3) USB-C/jack/PWR cutout alignment from §30 edge elevations, 4) front-side switch/slider/FSR-pad grid from the §2 front-panel layout. Everything else floats inside its zone.

This page + §31 (net list) together are the complete KiCad brief — schematic from §31, placement and routing strategy from here.